



Picking partners: Manager selection in private markets[☆]

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ABSTRACT

We study the selection of private market managers (GPs) for over 61,000 capital commitments by institutional investors (LPs) from a feasible opportunity set. LPs chase past performance but also seem surprisingly willing to invest in GPs without a track record: the probability that LPs select first-time or young GPs is quantitatively similar to GPs in the highest quartile of past performance. The most plausible explanation is that there is demand for exposure to private markets that is not fulfilled by incumbent GPs. The proclivity to invest in first-time or young GPs is not associated with higher future performance.

1. Introduction

Ivashina and Lerner (2018) report that the average allocations of US institutions to private markets increased from 7% in 2008 to almost 20% in 2017, and practitioner sources often report higher and increasing allocations (e.g., [Clearwater Report](#)). Since the dollar value of individual capital commitments to each fund is generally small, fulfilling target allocations requires a large number of commitments. This frequency, combined with high investment costs per dollar invested ([Andonov, 2024](#)) means that Limited Partners (LP) spend a significant amount of time and resources on due diligence. Selection in private markets is especially challenging because, unlike public markets, the prior performance of General Partners (GP) raising funds is often not available. On the one hand, it has long been conventional wisdom

to avoid first-time funds without a performance history ([Harris et al., 2023](#)), and [Lerner and Leamon \(2023, pg. 49\)](#) point out that LPs are “keenly interested in track records because they are investing in a blind pool with long holding periods”. On the other hand, [Swensen](#) argues that “track records are really overrated” and that “some of Yale’s best investments have been with people that don’t have a track record”. In this paper, we study how LPs select funds, but with a particular focus on those raised by GPs for whom prior performance is unobservable.

Selection is typically analyzed using discrete choice models, in which agents choose from a set of available options. Estimation machinery pioneered by [McFadden \(1974\)](#) requires the choice set to be fully observable.¹ Unobservability of the choice set creates an identification issue because observed choices are functions of both preferences and the choice set. To resolve this difficulty, researchers (a) assume that

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¹ If private equity is a competitive market, as argued by [Cochrane \(2005\)](#), all funds raising capital should be available for investment to all LPs. However, if elite GPs ration access, selection methods must account for this constraint. A fuller discussion follows in Section 4.1.

² The literature that considers these approaches spans economics, finance, and marketing. [Barseghyan et al. \(2021\)](#) discuss heterogeneous choice sets (see also [Crawford et al., 2021](#)). Two studies in private equity use choice sets, but neither is directly concerned with selection in a broad sense. [Cavagnaro et al. \(2019\)](#) use a bootstrap to investigate the issue of luck versus skill. [Barber et al. \(2021\)](#) use a discrete choice model in a random utility framework to investigate willingness-to-pay for impact funds.

the choice set includes all feasible alternatives, or (b) use randomized choice sets, or (c) use supplementary information to create a choice set from a larger universal set of alternatives.² As described by Barseghyan et al. (2021), the last approach uses two-way exclusion restrictions via variables that affect the choice set but not preferences (and vice versa), and institutional details to model how variables affects the opportunity set (Gaynor et al., 2016).

Taking this approach, we study the decisions of 2692 North American LPs who make 61,602 capital commitments between 1995 and 2019. These commitments are made to a variety of fund types within the private markets umbrella, including buyouts, venture capital, growth, distressed debt, mezzanine financing, real estate, natural resources, and infrastructure. We take the perspective that LPs select a fund from GPs raising capital within a window around LPs' actual commitments. This superset includes failed fundraising attempts by GPs and represents the starting point of our analysis.

From this superset, we enforce five types of restrictions and allowances to generate investable choice sets, hereafter referred to as opportunity sets. These restrictions are informed by (a) the existing literature discussed later in the paper, (b) interviews with three large LPs, and (c) investment policy statements of 44 LPs (culled from 100 LPs in our sample with the most capital commitments). Deferring the details, the opportunity sets account for (a) heterogeneity in LPs' access to elite GPs, (b) acceptable commitment sizes from LP and GP perspectives, (c) heterogeneity in allocations to subcategories within the private markets umbrella, (d) differences in LPs' flexibility with respect to the timing of allocations, and (e) personal connections between LPs and GPs. Access restrictions implement the idea that certain GPs may ration and not accept capital from some LPs. Commitment size restrictions recognize that selection is a multidimensional decision involving not just selecting a fund but also determining the size of the stake; in doing so, we enforce restrictions based on minimum and maximum commitment sizes derived from LP and GP investment history. Heterogeneity in fund type reflects the notion that large LPs may strategically choose exposure to various types of private markets at different times. Timing flexibility permits large LPs to have intertemporal latitude in choice; for example, waiting much longer to invest in the funds of preferred GPs rather than choosing among GPs currently raising capital. Finally, personal connections acknowledge the importance of relationship-based fundraising in private markets.

Armed with these LP-year-specific opportunity sets, we estimate choice models. The coefficients on the explanatory variables represent the probability of selecting a fund of a GP with a particular attribute; the probability of selection is higher if the attribute is more common among selected funds relative to funds in the opportunity set. Comparisons of these probabilities across attributes forms the centerpiece of our analysis. Since our interest is in GP-level attributes relevant for selection, the verbiage that we use in most of the paper is 'GP selection,' even though it is a fund that is being selected.

Prior performance is potentially an important determinant of manager selection. Since funds currently raising capital do not have a performance history, LPs rely on the performance of prior funds raised by the GP. Our regressions use the asset-weighted average Internal Rate of Return (IRR) of prior funds raised by GPs to classify them into performance quartiles. Consistent with performance chasing, there is a robust monotonic relation between prior performance and the probability of selection.

Perhaps more surprising are the large selection probabilities of funds raised by GPs that have no or limited performance history. We separate such GPs into two groups: first-time GPs that are raising capital in their first fund, and young GPs that are raising capital for subsequent funds but have not been around long enough to generate reliable measures of fund-level performance. The incremental probability of selecting a first-time GP is as large as that of a GP in highest past performance quartile (1.06% compared to 1.02%, both relative to the GPs in the lowest quartile of past performance). The incremental

probability that a young GP is selected is also relatively large (0.76%). The fact that GPs without observable performance are as likely to be selected as GPs in the highest past performance quartile is quite surprising. A battery of tests reveals that the tendency to invest in first-time and young GPs is robust to alternative ways of constructing the opportunity set.

We add color to this result in four ways. First, we examine the role of intermediaries in selecting GPs, with a particular focus on their influence in selecting first-time GPs. Both demand-side intermediaries, such as search consultants used by LPs (Andonov et al., 2025; Martinez and Qian, 2025), and supply-side intermediaries, such as placement agents used by GPs (Cain et al., 2020), increase the probability of selecting first-time GPs but only modestly so. Second, using hand-collected information on the origin story of first-time GPs, we separate them into "veteran" GPs whose founders were previously employed in private markets or the financial sector, versus "rookie" GPs that had no such prior experience. The relative selection probabilities for both rookie and veteran GPs are high, although they are marginally higher when the founder of a first-time GP acquired prior experience in a large well-established GP. Third, we examine whether personal connections between LPs and GPs are related to the probability of selection. Personal connections significantly increase the probability of selection across all GPs (by 0.54%), but the effect is notably stronger for first-time GPs (0.96%). Fourth, we investigate variation across LPs, such as differences in decision-making process, scale economies, experience in private markets, and organizational type.³ The tendency to select first-time GPs is stronger among larger LPs, public pension systems, LPs with formal investment committees, and LPs making their first private market investments. This richness and variation reinforces the perspective that LPs are not monolithic in their decision-making. Across all these tests, the probability to select first-time GPs relative to GPs in the highest past performance quartile always remains high.

Given this pervasiveness, it is natural to ask why LPs tend to select first-time GPs. We consider several plausible explanations. One possibility is that LPs believe that the expected returns from first-time GPs are high. Instead, we find that excess expected IRRs (using the methodology of Barber et al., 2021) are -1.65% for first-time funds, compared to 4.45% for funds raised by GPs in the highest quartile.⁴ And these expected returns align closely with ex post realized IRRs: the average excess realized IRR for first-time funds is -1.08% but is 2.63% for funds raised by GPs in the highest quartile. Another possibility is that LPs have a lottery preference, inasmuch as first-time GPs might generate skewed payoffs. However, we do not observe higher skewness in the performance of first-time GPs, especially compared to funds launched by GPs in the highest past performance quartile. We also consider the possibility that LPs invest in first-time GPs to establish a toehold in GPs that subsequently become star performers and ration access. There is no evidence to support such an explanation: the number of subsequent funds raised by first-time GPs that end up as top quartile GPs is small, particularly relative to the number of first-time funds originally created. In addition, LPs' commitments to follow-on funds of first-time GPs are not associated with superior performance. Finally, first-time GPs may attract LPs by disproportionately offering co-investment opportunities that lower overall fees (Braun et al., 2020; Fang et al., 2015). But for a subsample for which we can observe such data, we find no evidence that first-time GPs are more likely to offer co-investment opportunities.

³ For example, Lerner et al. (2007) document the superior performance of endowments which they attribute to information acquired as inside investors in reinvestment decisions. Sensoy et al. (2014) argue that this superior performance is due to preferential access to early high-performing VC funds. Da Rin and Phalippou (2017) suggest that endowments are not special but that LP size is related to due diligence and investment.

⁴ Explicitly including expected IRRs in the selection equations does not reduce the selection probabilities of first-time GPs relative to that of GPs in the highest past performance quartile.

Occam's razor suggests a simple demand–supply explanation in which LPs that face pressure to invest gravitate towards first-time GPs relative to GPs in the highest past performance quartile. To evaluate this possibility, we construct a measure of pressure-to-invest in a given fund type by combining two sources of information (a) deviations of actual portfolio weights from target weights, and (b) growth in allocations. The former is available for 347 LPs while the latter can be calculated for all LPs. Using this variable, we find that LPs facing high pressure to invest are more likely to select first-time GPs than GPs in the highest past performance quartile. In contrast, LPs under no pressure to invest are *less* likely to select first-time GPs than GPs in the highest past performance quartile. This implies that the main mechanism that is driving the surprising tendency to select first-time GPs is the pressure to invest. These selection propensities, combined with the fact that the expected returns of first-time GPs are lower than those of GPs in the highest past performance quartile, suggests that LPs facing high pressure to invest navigate down the performance ladder.

Although our main purpose is documenting and explaining the surprisingly large tendency of LPs to invest in first-time GPs, we also study other factors that play a role in selection: local bias (Hochberg and Rauh, 2013), reinvestment decisions (Hochberg et al., 2014; Lerner et al., 2007), and peer effects such as following the Yale Model (Swensen, 2000). Our contribution regarding these factors is twofold. First, we quantify the selection probabilities of GPs associated with these factors, which can only be estimated by constructing an appropriate opportunity set. Reinvestment decisions are especially important: a GP in which an LP has previously invested is 9.7% more likely to be selected than one that has not. The incremental selection probability for local (in-state) GPs compared to non-local GPs is 0.98%. Second, we find that LPs tend to follow their peers' prior investment choices, with GPs previously selected by peers being 0.58% more likely to be chosen. This peer effect is not driven by GP reputation and, to our knowledge, is a novel finding.

The enormous growth of private markets as an asset class in institutional portfolios begs the question: Where does the capital go? The question takes on added importance with private equity making inroads into retail investor portfolios.⁵ From a market clearing perspective, as institutional investors raise their demand for exposure to private markets, this demand has to be absorbed by increases in funds sizes of incumbent GPs, or an increase in alternative investment vehicles outside of mainstream funds, or the entry of new GPs, or some combination thereof. Harris et al. (2023), and Kaplan and Schoar (2005), and others show that many high-performing GPs do not disproportionately increase their fund sizes. Information frictions and organizational constraints may explain why incumbent GPs not only refrain from increasing fund size (Lopez-de Silanes et al., 2015; Marquez et al., 2015) but also fees (Hochberg et al., 2014), leaving excess investor demand to be absorbed through other channels. Lerner et al. (2022) show that alternative vehicles play an increasingly important role in absorbing some of this demand. Our results imply that entry by first-time GPs is also an important component of the market clearing equilibrium. In fact, in our sample, first-time and young GPs together attract 25,942 commitments totaling \$715 billion.

The remainder of the paper is organized as follows. Section 2 provides a brief overview of the investment process with respect to private markets and includes a discussion that motivate variable choices.

⁵ Registered Investment Advisors (RIAs) increasingly allocate the assets of wealthy individual investors to private equity, often via interval funds (<https://www.wealthforge.com/insights/allocations-to-alternatives-results-from-our-ria-survey>). Even firms that cater directly to individual investors, such as Vanguard, partner with GPs to allow qualified individual investors to invest in private equity (<https://www.morningstar.com/articles/1041026/vanguard-steps-further-into-private-equity>). Newly issued rules by the Department of Labor also permit 401K plans to invest in private equity (<https://www.natlawreview.com/article/dol-creates-path-401k-plans-to-offer-private-equity-investment-options>).

Section 3 describes the data and an overview of the sample. Section 4 contains the main selection regressions and Section 5 examines explanations for commitments to first-time GPs. We conclude in Section 6.

2. The investment process

In this section, we first briefly describe the investment process of institutional investors in private markets, and then turn our attention to the economic drivers of selection.

2.1. LP and GP perspectives

The asset allocation process at most LPs designates a target weight (sometimes with lower and upper bounds) to private markets to be achieved over annual or multi-year periods. In large public pension systems, investment policy statements (IPs) disclose asset allocation processes, descriptions of what the institution regards as an independent asset class, target weights, the nature of the selection/approval process, ongoing monitoring, etc. IPs are not standardized, and there is considerable variation in the information provided about search and due diligence processes. For large public pensions, the details of the search and selection process sometimes appear in the public domain via request for proposals and meeting disclosures. For corporate pension systems and private endowments, the process and deliberations are kept confidential. We employ two approaches to develop insight into the process and to guide our empirical tests. First, we take the top 100 LPs in our sample with the most capital commitments and download IPs for those that are publicly available. We are able to obtain IPs for 44 LPs, all of which are public pension systems. We then read these statements with an eye towards understanding the allocation and selection process. Second, we interview the Chief Investment Officers of one large public pension system, one corporate pension system, and one university endowment to obtain contextual information on the nature of their processes. We also have access to the investment committee and policies of a large non-profit foundation that is active in this market.

Although the information we obtain does not constitute large sample empirical evidence, it guides downstream empirical procedures in several ways. First, we observe heterogeneity in the consideration of private markets and investment focus; some LPs focus on only a few private markets, others invest across all. For example, some LPs only consider private equity, while others lump private equity together with real estate, private credit, real assets (including infrastructure and natural resources) as one “private markets” asset class. Some of this variation seems to be tied to the size of the LP: large LPs are more willing to consider investing across private market categories and over time. Second, to manage risk, LPs assess their investment capacity in a fund, both in absolute terms and relative to the target fund size. Third, some LPs, particularly those that are large, have formal investment committee structures, while others do not. Fourth, many LPs use investment consultants, both in their general processes such as asset allocation and asset-liability management, and in targeted searches for GPs. Search consultants used in private markets are often not the same as general consultants and serve as intermediaries that can address specialized information, due diligence, and access issues (see Andonov et al., 2025; Begenau et al., 2024; Martinez and Qian, 2025).

The above narrative reflects an LP's perspective. It is equally important to consider the GP's perspective, as it can influence the opportunity set available to LPs. For instance, it is often argued that elite GPs may only permit access to LPs who have previously invested in their funds, as well as to LPs with sizeable portfolios and deep pockets, thereby limiting the investment opportunities available to other LPs (see, for example, discussions and evidence in Hochberg et al., 2014; Kaplan and Schoar, 2005; Lerner and Leamon, 2023; Lerner et al., 2022, 2007; Sensoy et al., 2014). In contrast, some GPs employ intermediaries (known

as Placement Agents) to expand the set of potential investors; Cain et al. (2020) document evidence consistent with both influence peddling and certification motives. Due to SEC regulations that prohibit public solicitation for unregistered securities offerings, GPs may also rely on personal networks and informal channels to attract capital (Howell et al., 2025). Personal relationships may be especially important for small GPs or GPs lacking an established performance history.

We integrate the insights from these perspectives into our empirical design in Section 4.2 but for now proceed to a discussion of plausible economic drivers. The rubric we employ separates these drivers into informational versus non-informational channels. Some of the variables that we categorize as information-related can also be non-informational in nature, but we keep the two channels separate for expositional ease.

2.2. Economic drivers

2.2.1. Informational channels

LPs' line-of-sight into portfolio companies and their valuations is limited by imperfect disclosure and lack of market prices.⁶ The informational asymmetry created by this opacity generates search costs. LPs use a variety of mechanisms to manage informational problems, some of which have been studied in the existing literature, although typically in the context of LPs' performance rather than selection. We discuss these mechanisms below, motivating their use in subsequent regressions.

- Past performance or lack thereof: In summarizing the literature on performance persistence in private equity, Harris et al. (2023) describe the conventional wisdom for investors in private equity to choose funds run by GPs with high past performance, while avoiding first-time funds. If an LP believes there is persistence in performance, it is easy to see how chasing performance is relevant for GP selection. Even in the absence of persistence, however, the mere existence of a prior fund generates information about the investment capabilities of a GP. First-time GPs, therefore, likely face higher hurdles in raising capital. Some informational issues might be mitigated via accumulated industry-specific human capital. Therefore, it is possible that industry veterans who depart from well-known GPs to create their own firms are able to persuade LPs to commit capital by virtue of their employment history. On the other hand, Lerner and Leamon (2023, pg. 49) write that "LPs are notoriously unwilling to fund new groups, even groups spinning out on their own after working in a larger organization". The null hypotheses that appear from these arguments is that selection is positively related to past performance, and that selection probabilities for first-time GPs should be lower than those for GPs with good past performance.⁷
- Prior investment: Investing in a prior fund provides an LP with information rights and allows it to exert some amount of governance pressure on the GP (Hochberg et al., 2014; Lerner and Schoar, 2004) and influence reinvestment decisions (Lerner et al., 2007). The insights gained from past investments may also cultivate trust, and limit search and due diligence costs, including the learning channel described by Phalippou and Gottschalg (2009). Therefore, we expect prior investment to be positively related to selection.

⁶ Recently adopted SEC rules and amendments for private funds are designed to address such informational asymmetries, such as requiring that GPs send quarterly financial statements, conduct annual audits, disclose side letters, and provide fairness opinions (<https://www.sec.gov/newsroom/sec-stories/sec-approves-private-fund-adviser-reforms>).

⁷ Interestingly, industry publications tout the benefits of investing in first-time GPs and even argue that their performance can exceed that of well-established GPs (Preqin, 2016). We show later in Section 5.1 that there is no evidence of excess performance of first-time GPs.

- Local information: Hochberg and Rauh (2013) argue that when disclosure is scarce and information acquisition is expensive, local networks and connections can be valuable. If that is the case, the probability of selecting a fund sponsored by a local GP should be higher than that for a non-local GP.
- Peer choices: LPs participate in a variety of industry conferences, often marketed as networking events to meet peers as well as GPs. In addition, LPs often use peer group universes to benchmark themselves against each other and University endowments use the data published by the National Association of College and University Business Officers (NACUBO) for comparisons.⁸ It is also possible that university endowments follow their peer group under the so-called Yale model pioneered by Swensen and Taka-hashi (Swensen, 2000). A capital commitment (or lack thereof) by a peer LP may therefore contain information similar to a reinvestment decision, implying that selection probabilities may be positively related to prior selections by peers.⁹

2.2.2. Non-informational channels

Non-informational reasons can also be important for selection. There can be variation in fees across funds (Robinson and Sensoy, 2013), and within funds (Begenau and Siriwardane, 2024). One way in which LPs lower investment costs is through co-investment programs in which LPs invest directly in portfolio companies (Braun et al., 2020; Fang et al., 2015) or other alternative vehicles (Lerner et al., 2022). Since LPs pay no (or lower) fees and carry in direct investments, overall investment costs are lower. Finally, LPs can have nonpecuniary objectives in selecting GPs; some public pension systems have "Emerging Manager Programs" or programs dedicated to minority-owned firms.

2.3. LP heterogeneity

LPs differ substantially in terms of investment expertise, scale economies, governance structures, and investment experience in private markets. It is often argued that these differences show up in performance differentials. Lerner et al. (2007) report large differences in the performance of private equity investments of various types of LPs and note the superior performance of endowments. Sensoy et al. (2014) argue that this superior performance is due to preferential access to high performing venture funds specific to the 1991–1998 sample period. Cavagnaro et al. (2019) report large differences in skill between LP types. Independent of LP type, the size of an LP and its prior experience in private markets are also likely to be tied to the selection process. Small LPs that lack scale economies are less likely to be able to afford the human capital required for the due diligence process (Da Rin and Phalippou, 2017). Similarly, LPs who have no prior experience in private markets (versus those who make repeated allocations) are likely to face higher search costs. The upshot of all of this is that there is likely considerable heterogeneity in the selection process and preferences across LPs.

3. Data, sample construction, and capital allocations

3.1. Data sources and sample construction

We use several customized data files provided by Preqin. A capital commitment file records allocations made by LPs along with the vintage year of the fund, fund type, fund region representing the geographic

⁸ See, for example, <https://www.bnymellon.com/us/en/solutions/securities-services/peer-group-universes.html>.

⁹ There are also other non-informational motivations for following a peer. For example, Bursztyn et al. (2014) suggest that following peers might simply generate social utility, similar in spirit to a reference dependent utility function such as "keeping up with the Joneses".

focus of the fund, the target and final fund size, the domicile of the LP, and LP type. A performance file contains a time series of performance information for all private market funds in the Preqin universe, GP name, and domicile. Other files contain LPs' assets under management, information on whether a fund permits co-investment by LPs, and whether a fund used placement agents. To ensure that backward-looking measures such as prior performance are not left-truncated, we start the sample in 1995. The sample ends in 2019.

Brown et al. (2015) describe Preqin's data collection methods in considerable detail, so we do not duplicate them here except to note three relevant aspects (see also Begenau et al., 2020). First, some LPs and top-tier GPs may be less willing to disclose information. To the extent that non-reporting is concentrated in private LPs, we replicate our tests across different types of LPs to ensure that selective disclosure does not bias our results.¹⁰ With respect to non-reporting by top-tier GPs, Brown et al. (2015) conclude that the four commonly used databases (Burgiss, Cambridge Associates, Pitchbook, and Preqin) are more comparable within North America, are similar for buyout funds, but with some discrepancies in venture capital. We ensure that our results are not driven by any specific fund type, including venture capital. Second, Preqin uses data from GPs and LPs to generate a fund-level IRR, giving preference to GP sourced data. If data are sourced from multiple LPs, then Preqin uses the following order: first-close investor, timeliness of the source, and reliability of the source. To the extent that fund-level performance measures are drawn from a mix of high- and low-fee LPs, any bias is likely diversified away (Begenau and Siriwardane, 2024). Third, data fields can sometimes be missing. For example, of the 61,602 capital commitments that we examine, the dollar value of the commitment is available for 26,171 records. We retain records with missing commitment size information so as not to introduce unintended biases into the analysis. For a subset of tests, we impute missing commitment sizes.¹¹ A fuller discussion of robustness tied to missing or potentially mismeasured information is in Section 4.4.1.

From the capital commitment file, we eliminate obvious data errors and commitments to Multi-Manager Funds, Separate Accounts, Secondaries, and Miscellaneous funds. The bulk of our tests exclude Fund-of-Funds but in robustness tests, we include these as well. Although LP and GP coverage is global, we restrict our attention to North American LPs and GPs. The geographic focus of funds launched by North American GPs can be in other countries, but as an empirical matter, almost 92% of funds are North-American-focused. We exclude LPs classified by Preqin as private equity firms, fund-of-fund managers, infrastructure firms, hybrid firms, and hedge funds.

The capital commitment file is linked to performance via unique IDs. The net IRR for each fund-quarter is calculated by Preqin as the money-weighted return using the present value of contributed and distributed cash, and the fair market value of unrealized investments. The database also includes Total Value to Paid-In Capital (TVPI) which we use in robustness tests. Both IRR and TVPI are net of fees.

Because the Preqin database only contains data from funds that were successful in raising capital, we assemble a data set of failed funds from three additional sources. First, we obtain data on buyout, venture,

and growth funds that failed to raise capital between 2009 and 2019 from the Institute for Private Capital (IPC). Second, we impute failed fund raising attempts for all fund types and years not included in the IPC data. If a GP has not raised a follow-up fund until the end of our sample, we assume that the GP attempted to raise a fund at a future date (we use the median time between consecutive funds in that fund type to determine the date of possible fundraising attempt). Third, to identify failed funds launched by first-time GPs that are not covered by IPC, we use the methods of Cassel et al. (2022) that rely on Form D filings with the SEC during the 2008–2019 period. In doing so, we keep funds defined as pooled investment funds but drop hedge funds, feeder funds, sidecars, secondaries, offshore funds, and coinvestment vehicles. Across the three methods, we identify 2549 failed fundraising attempts.¹²

Some of our tests require additional information on (a) the use of search consultants, (b) LPs' target weights, and (c) personal relationships between LPs and GPs. For search consultants, we use the Fundmap data in Goyal et al. (2024), similar to that in Martinez and Qian (2025). For target and actual weights, we obtain data from CEM Benchmarking, which collects these data directly from LPs (see Andonov, 2024 for details). For personal relationships, we use the RelSci dataset that contains information on links between employees of LPs and GPs (see the Appendix of Goyal et al., 2024 for details).

3.2. Origin and destination of capital

The sample contains 2692 unique LPs, which Preqin classifies into 40 different types, many of which are sparsely populated. To determine peer groups, we consolidate these into 10 groups that represent economically significant differences in structure, purpose, and governance.¹³ For reporting purposes and subsample analysis, we further consolidate these into four groups: corporate pension systems, public pension systems, endowments & foundations, and a miscellaneous group.

Table 1 provides information on the origin of capital commitments. The first set of columns shows the number and size distribution of LPs. Endowments & Foundations are the most populous, but are substantially smaller than the other groups. The data contain 405 public pension systems, including both large state-level systems and small city and municipality plans. The median size of public pension systems in our sample period is almost \$10 billion, and the 90th percentile is greater than \$79 billion. The miscellaneous group includes insurance companies, banks, and two sovereign wealth funds (Canada and Alaska) that skew the size distribution. The second set of columns shows the number of capital commitments for each LP group, the median dollar commitment, and the total amount of capital committed. Of the 61,602 capital commitments in our sample, almost 35% (21,310 commitments) originate from public pension systems. These systems also have the largest median commitment size (about \$30 million), totaling more than \$1.2 trillion.

Table 2 describes the destination of capital. Panel A shows statistics on funds that successfully raised capital, organized by the type of fund. The first set of columns shows the number of funds in each category and the distribution of fund size. Of the 7939 funds in this sample, 4406 are traditional private equity funds (buyout, growth, and venture). Real

¹⁰ Preqin informs us that they also scrape Form ADV filings, in addition to using FOIA and other data sources.

¹¹ In doing so, we use the following hierarchy: (a) we use the median commitment size that the LP committed to other funds in that fund type in the same year, (b) we use the median commitment size committed by LPs in the same size decile for that fund-type-year, separately for first-time and non-first-time GPs, (c) if medians in (b) are not calculable, we use the median commitment size committed by LPs in the same size decile for that fund-type-year, (d) if medians in (c) are not calculable, we use the median commitment size committed by LPs in the same size decile and fund-type. The vast majority of the imputation (88%) takes place using (b). Our main selection regression do not rely on imputed values.

¹² Our sample of failed fundraising attempts does not include independent sponsors who may raise capital on a transaction-by-transaction basis rather than the traditional blind-pool structure. Independent sponsors are not registered with the SEC, we cannot track their activity, and they do not appear in the databases available to us.

¹³ The group of 10 include corporate defined benefit plans, private foundations, public foundations, private universities, public universities, unions, state-level pension systems, local (non-state) pension systems, sovereign wealth funds (the Alaska Permanent Fund and the Canadian Public Pension Plan), and a miscellaneous group.

Table 1
Sources of capital.

	LPs				Commitments		
	N	AuM percentile (\$M)			N	Median \$M	Total \$B
		10th	50th	90th			
Corporate pensions (Corp. pensions)	420	931	5670	45,463	11,828	25	266
Public pensions (Publ. pensions)	405	713	9518	79,136	21,310	30	1217
Endowments & Foundations (E & F)	1106	90	678	6946	16,229	12	184
Miscellaneous (Misc.)	761	662	13,228	261,086	12,235	15	389
All LPs	2692	274	5062	80,200	61,602	25	2224

Public pension systems comprise state, municipality, city, and public worker pension systems. Endowments and Foundations include both public and private organizations. The miscellaneous group includes insurance companies, banks, and two sovereign wealth funds. The table reports the number of LPs and the 10th, 50th, and 90th percentile of their assets under management (AuM), in \$ million, at the end of the sample period for each LP. The table also reports the number of capital commitments, the median commitment size (in \$ million), and total value of all commitments (in \$ billion). Since the dollar value of capital commitments is only available for a subsample, total commitments include imputed values, as described in the text. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

Table 2
Destination of capital.

	Funds/GPs				Commitments			# commitments
	N	Percentile fund size (\$M)			N	Median \$M	Total \$B	per fund/GP
		10th	50th	90th				
Panel A: Fund type								
Buyout	1811	106	644	4000	19,330	28	764	23
Direct lending	273	200	670	2300	1121	43	62	16
Distressed PE	362	250	1000	3241	4499	30	157	33
Growth	565	100	390	1500	3474	23	94	17
Infrastructure	207	100	517	4500	2062	40	101	13
Mezzanine	341	90	261	1250	2575	17	66	15
Natural resources	360	150	500	2000	4282	25	119	20
Real estate	1990	100	400	1584	12,856	30	518	13
Venture	2030	25	180	700	11,403	12	174	15
Total	7939	75	407	2400	61,602	25	2056	16
Panel B: Past GP performance								
PastPerfLow	186	123	380	1265	1133	25	38	15
PastPerfQ2	457	100	477	2300	3866	28	140	17
PastPerfQ3	832	180	725	4250	10,439	30	433	24
PastPerfHigh	955	170	738	5000	14,261	30	562	25
FirstTime	2209	23	177	666	8479	16	206	11
Young	2583	70	307	1250	17,463	20	509	15
PastPerfMiss	717	91	340	1390	5961	20	168	17
Total	7939	75	407	2400	61,602	25	2056	16

The first column report the number of funds receiving commitments in our sample. The second, third and fourth columns report the 10th, 50th, and 90th percentile of distribution of fund size in \$ million for funds raised in the last 5 years of the sample period (2015–2019). The fifth, sixth and seventh columns report number of capital commitments, the median commitment size (in \$ million), and total value of all commitments (in \$ billion), respectively. The total value of commitments includes imputed values, as described in the text. The last column shows # commitments per fund/GP, calculated as the ratio of median fund/GP size to the median commitment size. Panel A reports these statistics by fund type. Panel B reports them based on past GP performance at the time a fund receives a capital commitment. For GPs with available past IRRs, we rank GPs into performance quartiles using asset weighted past IRR each year. These quartiles are labeled PastPerfLow, PastPerfQ2, PastPerfQ3, and PastPerfHigh. GPs without past performance fall into three categories: (1) First-time GPs (labeled FirstTime) are those raising capital for their first fund. (2) Young GPs are those who are not identified as FirstTime but are raising subsequent funds in the first five years of their life. (3) PastPerfMiss consists of a residual group of GPs that are more than five years old but their performance data is missing in Preqin.

estate has 1990 funds, and the remainder is roughly evenly distributed between direct lending, distressed debt, infrastructure, mezzanine debt, and natural resources. The median fund size is \$407 million, with substantial variation between fund types. Almost 56% (34,207 out of 61,602) of capital commitments are allocated to traditional private equity funds, and in dollar terms, these commitments represent over 50% of aggregate commitments (\$1032 billion out of \$2056 billion).

Panel B contains similar statistics for funds based on the prior performance of sponsoring GPs. To measure past GP performance, we require that more than five years elapse after the vintage year of a fund (Chung et al., 2012), and then roll up fund-level performance to the GP level by computing the fund-size weighted performance of all funds raised by the GP prior to that year. For example, consider a fund of a GP that receives a capital commitment in 2008. At that

time, the GP has raised three funds I, II, and III with closing dates in 1998, 2000, and 2004, respectively. At the time of capital commitment, we use the performance of funds I and II to compute the weighted average performance of the GP. Armed with these GP-level performance measures for each year, we use annual breakpoints to sort all GPs raising capital into performance quartiles, labeled PastPerfLow through PastPerfHigh. There are three types of GPs that are not included in the past performance quartiles: (i) GPs that are launching their first funds, labeled FirstTime, (ii) GPs who are not identified as FirstTime but are raising subsequent funds in the first five years of their life, labeled Young, and (iii) a residual group of older GPs, for whom sufficient time has passed to observe the performance of prior funds but their performance data is missing in Preqin, labeled PastPerfMiss.

As might be expected, fund size increases with prior GP performance. For example, the median fund size in the PastPerfLow quartile is \$380 million, rising to \$738 million for PastPerfHigh.¹⁴ Unsurprisingly, the median size of first-time funds is small (\$177 million) and rises to \$307 million for funds launched by young GPs. The number and aggregate dollar value of commitments also increases with GP performance: GPs in the PastPerfLow quartile receive 1133 capital commitments totaling \$38 billion but GPs in the PastPerfHigh quartile receive 14,261 commitments amounting to \$562 billion. First-time GPs receive 8479 commitments but since the median dollar value of commitments to first-time GPs is \$16 million, the aggregate value of commitments to first-time GPs is about \$206 billion.

The last column of Table 2 contains the number of commitments per fund, computed by scaling the median fund size by the median commitment size. The number of commitments per fund for first-time and young GPs is 11 and 15, respectively, similar to that of GPs in the lowest performance quartile (15), and substantially lower than funds in the highest performance quartile (25).

4. GP selection

4.1. Modeling framework

If the supply of private market funds is competitive, even with some lag, then a well-established approach to modeling selection is a discrete choice model with a full opportunity set. However, if GPs also ration access, then one can also think of this market in a two-sided matching framework. There are two critical elements to modeling matching markets: (a) whether or not utility is transferable without loss between coalition members, referred to as transferable utility (TU) and non-transferable utility (NTU) in the literature, and (b) whether the matching between agents is one-to-one, many-to-one, or many-to-many.

Equilibrium concepts in the matching literature generally rely on individual rationality and matching stability (see surveys by Chade et al., 2017; Chiappori and Salanié, 2016). Like Sørensen (2007), our setting is NTU with many-to-many matching because an LP can choose many GPs, and a GP can choose many LPs.¹⁵ Echenique and Oviedo (2006) emphasize that while the theory and estimation technology associated with one-to-one and many-to-one matching are well developed, for a variety of technical reasons, that is not the case for many-to-many matching. Fox (2018) is able to estimate a many-to-many matching model but requires TU, and regardless, is computationally infeasible for our sample sizes. He et al. (2024) can generate identification and estimation with NTU but only for many-to-one matches and their approach also suffers from the curse of dimensionality.¹⁶ Despite the above challenges, aspects of their approach guide the implementation in the market we study. Specifically, they employ exclusion restrictions that function as “feasible-set shifters” which help to identify preferences. That approach parallels the selection literature that employs exclusion restrictions in which some variables impact the opportunity set but not preferences, and vice versa (see, for example, Gaynor et al., 2016; Goeree, 2008). This permits a way forward via a standard utility

framework. Following Barber et al. (2021), consider an LP_i investing in $GP/Fund_j$, with a utility function:

$$U_{ij} = f(X_{1j}, X_{2ij}, \gamma_i), \quad (1)$$

where X_{1j} are GP/Fund characteristics, X_{2ij} are LP-GP characteristics, and γ_i is a baseline utility of investing in private markets. Under well-known distribution assumptions for errors in the estimation equation, this maps to a discrete choice model of Y_{ij} (where $Y_{ij} = 1$ for a selected fund of a GP, and $Y_{ij} = 0$ for non-selected funds) that can be estimated via a conditional logit (McFadden, 1974). To estimate such a model, our challenge is to identify the set of GPs raising capital from which an LP could plausibly select funds while excluding GP/Funds that are not accessible. We proceed to such an empirical design below.

4.2. Empirical design

Since LPs may make more than one capital commitment in a year, we construct an opportunity set for each LP-year. We start by identifying all funds (including failed funds described in Section 3.1) with closing dates within ± 1 year window from the average closing date of selected funds. This is an unrestricted opportunity set on which we impose restrictions (and some allowances) guided by the discussion in Section 2.1. We outline these below and then discuss the intuition behind each of them. Before doing so, we note that all restrictions are over-ridden if an LP has an investment in a previous fund of the GP. This is because of the strong evidence for reinvestment (Lerner et al., 2007), and because practitioners inform us that investing in a prior fund removes barriers to entering subsequent funds. Some of our procedures require us to identify “large” versus “small” LPs. We use the 90th percentile in the distribution of annual assets under management as a breakpoint, but in Section 4.5 we consider alternative breakpoints, in addition to other procedural combinations.

1. Access restrictions

- (a) We limit LP access to larger GPs based on their respective size percentiles, with the first size threshold starting at 75th percentile (i.e., top quartile) and increasing in 5% increments. For example, LPs below the 75th size percentile are unable to access GPs above the 75th size percentile. Similarly, LPs below the 80th size percentile cannot access GPs above the 80th size percentile, and so on.
- (b) For small LPs, we exclude a GP from the opportunity set if the GP's prior fund was oversubscribed.

2. Commitment size restrictions

- (a) We exclude a GP from the opportunity set of an LP if (i) the maximum historical commitment of an LP is less than the GP's minimum historical commitment, or (ii) an LP's minimum historical commitment is greater than the maximum historical commitment of the GP.
- (b) We exclude a GP from the opportunity set if an LP's historical median commitment is larger than 13% of the target fund size.

3. Fund type restrictions and allowances

- (a) For large LPs the opportunity set includes all types of funds.
- (b) For small LPs the opportunity set only includes fund types that the LP has invested in the past or is investing in the current year.

4. Timing restrictions

¹⁴ Quartile assignments are at the GP (not fund) level, and since poorly performing GPs are less likely to raise new funds, the number of funds in each quartile is unequal.

¹⁵ Sørensen (2007) studies matching between VCs and portfolio companies. While acknowledging that portfolio companies receive funding from many VCs (i.e., the setting represents many-to-many matching), he uses a one-to-many framework for estimation because he focuses only on the lead investor. Our setting does not permit such a dimension reduction.

¹⁶ We thank Jeremy Fox and Shruti Sinha for a discussion of estimation feasibility and related issues.

- (a) The window for constructing the opportunity set begins at ± 1 year for LPs in the largest size decile and decreases by 20 days for successive size deciles, reaching ± 6 months for those in the smallest decile.

5. Personal relationship allowances

- (a) If a GP has a personal relationship with an LP, the commitment size restrictions described in 2(a) (ii) and 2(b) are not binding.

Restrictions 1(a) and 1(b) broadly limit smaller LPs from accessing larger GPs or previously oversubscribed GPs. This generally reflects practitioner descriptions of the way in which rationing works and follows the discussions in Hochberg et al. (2014), Kaplan and Schoar (2005), and Lerner et al. (2007). The semi-continuous function accommodates the idea that the choice set varies more-or-less continuously with LP and GP size.

Lerner and Leamon (2023) describe how minimum commitment sizes prevent GPs from raising funds from a large number of LPs and then shirking, and maximum commitment sizes prevent an LP from exercising undue influence over a fund. In addition, LPs that need to make large capital commitments are unlikely to consider investing in very small funds, particularly given due diligence and monitoring costs. GPs also prefer that LPs' commitments are not too small because each commitment creates incremental reporting, coordination, and compliance costs. Given these two-way tradeoffs, commitment size restriction 2(a) rules out extreme mismatches between LPs and GPs in terms of historical commitment size (for each fund type). Restriction 2(b) reflect LP-level investment policy rules that ensure that an LP does not become too large a partner in a fund.¹⁷

Fund type restrictions and allowances (3(a) and 3(b)) recognize that large LPs can be opportunistic with respect to investment opportunities, seeking exposure to private markets in fund types where they deem investment opportunities attractive. As an empirical matter, small LPs concentrate in a few fund types: in our data, 67% of the LPs in the smallest size decile invest in only one fund type throughout the entire sample period, and 91% of all LPs limit their investments to no more than three fund types.

The timing restrictions in 4(a) reflect the idea that larger and more sophisticated LPs likely have greater intertemporal flexibility and may strategically add or forsake capital commitments over time. As with access, the function permits the choice set to vary semi-continuously with LP size.

The allowance in 5(a) relaxes commitment size restrictions for small GPs if they have a personal relationship with the LP. This reflects the earlier discussion that personal networks and relationships are important to GPs' fundraising efforts, particularly for small and inexperienced firms with limited track records (Howell et al., 2025).

4.3. Opportunity set characteristics

Table 3 describes the composition of the opportunity set. The first row shows that on average there are four capital commitments in an LP-year. There is some small variation across types of LPs and, unsurprisingly, LPs in the largest size decile make more capital commitments per year (8) than small LPs (3). The second row shows that the average number of funds in the opportunity set is 369, which includes failed funds. The distribution of the opportunity set size is positively skewed, as large LPs investing across all fund types have a broader selection of funds. The averages also obscure large time series variation.

The number of GP-funds in the opportunity set starts at 97 in 1995, but increases over time, fluctuating with the cycles of private equity fundraising activity.

The next set of rows (labeled "By Performance") report the number of GP-funds in the opportunity set in each past performance quartile, first-time GPs, and young GPs. On average, there are 95 (159) first-time (young) GP-Funds in the opportunity set. The last set of rows shows the average number of funds in the opportunity set for decisions in which (a) the LP has previously committed capital to the GP (PriorInv), (b) a peer of the LP has previously committed capital to the GP (PeerInv), and (c) the LP and GP are headquartered in the same state (Local). On average, there are 14 GP-funds in the opportunity set in which the LP has previously invested, 21 that are headquartered in the same state as the LP, and 114 in which peers have invested in the past.

4.4. Selection regressions

Table 4 presents selection regressions. Suppressing time subscripts for clarity, the dependent variable, $Y_{i,j,g}$, is equal to one for fund j , investing in region r , of fund type f , raised by GP g , that is selected by LP i , and zero for non-selected funds in the opportunity set. The first set of independent variables focus on past GP performance, including indicator variables for first-time GPs, young GPs, and past performance quartiles. Variables are lagged by two years in the regressions to ensure that the past performance of GPs is observable at the time of LPs' decision. Other variables include earlier defined indicator variables for prior investment (PriorInv), peer investment (PeerInv), and local investments (Local). The selection regression is:

$$Y_{i,j,g} = \gamma_1 FirstTime_g + \gamma_2 Young_g + \gamma_3 PastPerfMiss_g + \gamma_4 [PastPerfQ2_g, PastPerfQ3_g, PastPerfHigh_g] + \gamma_5 PriorInv_{i,g} + \gamma_6 PeerInv_{i,g} + \gamma_7 Local_{i,g} + \gamma_8 LnGPSize_g + FE_i + FE_{fr} + e_{i,j,g}, \quad (2)$$

where the omitted performance category is *PastPerfLow_g*. Since our opportunity set includes failed funds, we can interpret the coefficients as selection probabilities. Although selection is that of a fund, our interest is in GP attributes relevant for selection. In the remainder of the paper, we therefore use the phrase "GP selection" in describing the results.

Before proceeding to regression results, it is useful to delineate which of the above variables adhere to the two-way exclusion restrictions discussed in Section 4.1. In our setup, FirstTime, Peer, Local, and past performance quartiles reflect preferences but are not used to construct the opportunity set. Other variables such as LP and GP maximum–minimum commitment sizes, oversubscription of prior funds, and timing flexibility are used to construct the opportunity set but not included in the regressions. Some variables are used to construct the opportunity set and are also estimated in the regressions, such as PriorInv and GP size. Such dual usage is not uncommon and does not hinder identification provided that the opportunity set is modeled directly (Gaynor et al., 2016). As in the literature, we leverage the institutional detail described in Section 2.1 in such modeling. For example, LPs who have invested with a GP in the past are modeled to have access to a GP's future funds.

Column (1) of Table 4 reports conditional logit regressions (McFadden, 1974). It includes LP $\times$ Year fixed effects to account for time-varying heterogeneity across LPs because the size of the opportunity set varies across LP-years. Standard errors are also clustered at the LP $\times$ Year level because that is the unit of decision-making in our analysis. The coefficients on each of the performance quartiles increase from low to high past performance, and the coefficients on FirstTime and Young are positive, implying higher selection probabilities relative to PastPerfLow. The variables associated with prior investments, peer effects, and local bias are also positively related to selection.

¹⁷ In our dataset, the 90th percentile of the ratio between the dollar commitment of LPs and the size of the fund is 13%, reflected in 2(b).

Table 3
Opportunity set statistics.

	All	LP type				LP size	
		Corp. pensions	E & F	Publ. pensions	Misc.	Small	Large
# Commitments	4	4	3	6	3	3	8
	Number of funds						
All	369	384	237	459	497	248	905
<i>By existence</i>							
Failed	90	88	56	123	119	59	226
Raised	279	296	181	337	379	189	679
<i>By performance</i>							
FirstTime	95	103	76	104	113	78	170
Young	159	169	108	192	207	115	354
PastPerfLow	8	8	4	12	13	4	27
PastPerfQ2	18	18	8	26	28	8	60
PastPerfQ3	26	25	11	39	42	11	94
PastPerfHigh	28	26	11	43	46	11	106
<i>By other</i>							
PriorInv	14	15	8	26	13	9	39
PeerInv	114	129	47	158	174	53	383
Local	21	22	15	26	27	14	52

We construct an opportunity set for each LP-year using the procedure described in Section 4.2. This table provides summary statistics for the opportunity sets. The first row (# commitments) is the average number of commitments made by an LP in an year. The rest of the table shows the average number of funds in the opportunity set. Failed funds are those that failed to raise capital (see Section 3.1 for details). Raised funds are the funds that successfully raised capital around the time of commitment. The performance categories are defined in Table 2. The other categories are defined as follows. PriorInv includes fund in which the LP had previously committed capital to the GP. PeerInv includes funds in which peer institutions previously committed capital to the GP. Local funds are those for which the LP and GP are headquartered in the same state (in the case of Canadian LPs, local funds correspond to Canadian GPs). We report statistics for the overall sample as well as for different types of LPs (as described in Table 1) and for LPs of different sizes. An LP is defined as Large if its assets under management are in the top decile and Small otherwise. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

In conditional logit regressions with fixed effects, marginal effects are not calculable because the fixed effects required to calculate marginal effects are not actually estimated (see, e.g., Katz, 2001; Stammann et al., 2016).¹⁸ Given that, as well as the fact that conditional logit models are not estimable with a large number of fixed effects, we follow the recommendation of Angrist and Pischke (2009, pg. 197) and use OLS. To verify that there is no distortion in inferences from linear models, column (2) of Table 4 presents a specification with fixed effects and clustering identical to the conditional logit but estimated via OLS. As before, there is a monotonic increase in selection probabilities from low to high past performance quartiles and we observe precisely the same patterns in coefficients on other variables.

OLS also has interpretive advantages. Since GPs in the lowest past performance quartile are the omitted category, the coefficients of other performance quartiles represent incremental probabilities relative to the omitted group. The coefficients of other mutually exclusive indicator variables can be interpreted as normally and can be compared to each other. That is especially useful because it allows us to interpret the coefficient on FirstTime relative to that on a benchmark group. A natural benchmark might be the mean or median GP in terms of past performance because one might expect new entrants to default to the average (or median). However, to generate even starker comparisons, we compare the coefficient on FirstTime to that on PastPerfHigh throughout the paper.¹⁹

Variation in selection probabilities could come from aggregate changes in investment opportunities across fund types, region, and time (e.g., procyclicality in venture capital). In column (3) of Table 4, we add FundRegion $\times$ FundType $\times$ Year fixed effects to account for this

variation. Since a fund may appear in the opportunity set of multiple LPs, this requires that at the minimum the error terms be clustered at the fund level. We choose to be more conservative and cluster at the FundRegion $\times$ FundType $\times$ Year level, in addition to LP $\times$ Year.

Our primary interest is in the selection probabilities associated with the lack of visibility into past performance. The coefficient for FirstTime implies that first-time GPs are just as likely to be selected as GPs in the highest past performance quartile (the coefficients are 1.06% and 1.02% respectively and statistically indistinguishable from each other). Such a high selection probability for first-time GPs is counter to the conventional wisdom that LPs avoid GPs with no prior performance history. The coefficient on young GPs is also quite large (0.76%), about 75 percent that of the coefficient on GPs in the highest past performance quartile.

Selection in private markets is sometimes facilitated by intermediaries acting on behalf of LPs or GPs and it is plausible that these intermediaries affect the selection of first-time or young GPs. LPs may use search consultants who specialize in the asset class to alleviate access restrictions and/or serve as a screening function.²⁰ In column (4) of Table 4, we augment the baseline regression with an indicator variable (labeled Cons) equal to one if a search consultant is used. If a search consultant is not used or if the data are missing, the indicator variable is equal to zero. The use of search consultants increases the probability of selection across all GPs by 0.72%, and there is weak evidence that this effect is stronger for first-time GPs, as indicated by an interaction effect of 0.16% (t -statistic = 1.66). On the supply side, GPs may use placement agents to market funds (Cain et al., 2020). In column (5), we augment the baseline regressions with an indicator

¹⁸ Statistical packages such as Stata do deliver marginal effects but, in doing so, set fixed effects to zero. Fixed effects are likely to be non-zero in our setting because opportunity sets vary across LPs and over time.

¹⁹ The differences in the selection probabilities can also be put into economic perspective by comparing them with the unconditional probability of selection (which is 1% = 61,602/6,179,52).

²⁰ General investment consultants are used for a variety of functions such as asset allocation, reporting, and to guard against headline risk. Our interest is in the role of specialized search consultants in selecting first-time or young GPs, who are often not the same as general consultants. Andonov et al. (2025) study consultant turnover and Martinez and Qian (2025) study consultant network effects and performance in private markets.

Table 4
Selection regressions.

	Clogit	OLS			
		(2)	(3)	Intermediary =	
	(1)			Cons (4)	PA (5)
FirstTime	1.46 (35.88)	1.15 (38.56)	1.06 (10.32)	1.06 (10.30)	1.09 (10.48)
Young	0.70 (19.01)	0.86 (30.03)	0.76 (7.62)	0.76 (7.62)	0.74 (7.56)
PastPerfMiss	0.65 (17.25)	0.84 (27.39)	0.71 (6.15)	0.71 (6.17)	0.68 (5.96)
PastPerfQ2	0.41 (10.78)	0.39 (11.88)	0.30 (2.35)	0.30 (2.36)	0.27 (2.17)
PastPerfQ3	0.27 (7.47)	0.54 (15.31)	0.49 (4.07)	0.49 (4.09)	0.43 (3.55)
PastPerfHigh	0.58 (15.86)	1.08 (27.56)	1.02 (6.88)	1.02 (6.89)	0.99 (6.69)
PriorInv	1.47 (60.68)	9.54 (89.67)	9.71 (24.21)	9.70 (24.19)	9.72 (24.25)
PeerInv	1.18 (66.59)	0.60 (47.54)	0.58 (15.94)	0.58 (15.88)	0.55 (15.48)
Local	0.81 (43.39)	0.99 (30.54)	0.98 (17.06)	0.97 (17.00)	0.97 (17.02)
LnGPSize	0.59 (91.21)	0.42 (69.63)	0.50 (12.62)	0.50 (12.62)	0.47 (12.13)
Intermediary	–	–	–	0.72 (12.18)	0.91 (12.25)
FirstTime × Intermediary	–	–	–	0.16 (1.66)	–0.69 (–8.03)
FE	a	a	b	b	b

This table presents results of conditional logit (Clogit) or OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise. Most of the independent variables are defined in [Tables 2 and 3](#). There are two additional independent variables representing intermediaries: (1) Cons is an indicator variable equal to one if the capital commitment by the LP used a search consultant, and zero otherwise. (2) PA is an indicator variable equal to one if the GP used a placement agent, and zero otherwise. Fixed effects in columns (1) and (2) labeled ‘a’ are LP × Year; and in columns (3) to (5) labeled ‘b’ are LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All OLS coefficients are multiplied by 100. The number of observations is 6,179,652 corresponding to an unconditional probability of selection of 1% (61,602/6,179,652). The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

variable equal to one if the GP uses a placement agent (labeled PA) and its interaction with FirstTime. The use of a placement agent by a first-time GP increases the probability that the GP is selected by only 0.22% (0.91–0.69, *t*-statistic = 3.92). In summary, the selection probabilities of first-time GPs remain high, and increase marginally when intermediaries are involved.

With respect to the non-performance variables, the coefficient on PeerInv implies that LPs are 0.58% more likely to invest in GPs who have previously received capital from their peers, than in GPs who have not. To the best of our knowledge, this is a hitherto undocumented result. We also control for the possibility that this result is driven by time varying GP characteristics such as reputation. If prestigious GPs are more likely to be selected, they will have longer histories and more LP investments, linking reputation to both selection and prior peer investment. To control for this possibility, we add GP × Year fixed effects, comparing LPs with and without prior peer investment with the same GP. The results in column (1) of [Table A.1](#) indicate minimal change in selection probabilities, suggesting that reputation effects do not drive selection. In addition, consistent with [Hochberg and Rauh \(2013\)](#), there is evidence of local bias (the coefficient for Local is 0.98%). Prior investment is the largest determinant of selection (9.71%), consistent with the high reinvestment rates documented by [Kaplan and Schoar \(2005\)](#).

In summary, the key result from our perspective is the surprisingly large willingness to select funds of first-time and young GPs relative to those of GPs in the highest past performance quartile.

4.4.1. Robustness, missing variables, and sample selection

Given the centrality of this result, we subject it to a series of robustness tests. Our regressions include a large number of fixed effects to control for differences in the unconditional probability of selection for each LP over time, as well as time-varying differences across fund types and geographies. Columns (2) to (4) of [Table A.1](#) presents regressions without fixed effects that address concerns about oversaturation, with alternative sets of fixed effects, and with more conservative clustering. In each of these specifications, the coefficients on FirstTime and Young remain elevated relative to that on PastPerfHigh. Additionally, in [Table A.2](#), we use the same specifications as [Table 4](#) but measure performance using TVPI instead of IRR ([Phalippou, 2020](#)). In the baseline specification of column (3), the coefficients on FirstTime and Young are 0.95% and 0.65%, similar to the coefficient for PastPerfHigh (0.78%).

It is also possible that instead of the coefficient on FirstTime being high relative to PastPerfHigh, it is the coefficient on PastPerfHigh that is low. This could happen if LPs no longer engage in performance chasing towards the end of our sample period, given the decline in performance persistence over time ([Harris et al., 2023](#)) and the maturation of the industry ([Lerner et al., 2022](#)). However, when we split the sample into two subperiods (1995–2007 and 2008–2019), the coefficients on FirstTime and PastPerfHigh remain comparable. In addition, 10-year rolling regressions similarly show that coefficients of FirstTime hover close to those of PastPerfHigh throughout the entire time period. This suggests that the predilection that we highlight is not transient or tied to changes in performance chasing.

Our sample also includes failed fundraising attempts derived from three sources: IPC data, imputed funds, and Form D filings. One may be concerned that the selection probabilities in Table 4 are driven by those sources. We re-estimate regression of column (3) in Table 4 removing one source at a time. Columns (2) to (4) of Table A.3 show that in every regression, the coefficients on FirstTime remain high relative to those on PastPerfHigh.

We do not observe all fund-level characteristics including soft information as well as hard characteristics that LPs may be privy to in offering documents such as LP rights and fees. It is possible that these unobservables represent important decision criteria that are absent in our selection model. For some variables that we do observe, the data are incomplete. We discuss these data issues below.

There are 1226 funds in the opportunity set which do not receive any commitments from the LPs in our sample. Of these, 553 funds receive commitments from non-North American LPs, LPs that Prequin classifies as private equity firms, fund-of-fund managers, infrastructure firms, hybrid firms, and hedge funds. The remaining 673 funds are extraordinarily small funds with median size of \$40 million (compared to a median size of \$407 million in the remaining sample), over 40% of these are real estate funds, and about 40% are first-time funds. The fact that these first-time funds never get selected by our sample of LPs attenuates the coefficients on FirstTime, making the selection probabilities of first-time GPs conservative.

Performance data for some funds may be non-randomly missing or mismeasured relative to that observable by LPs at the time of decision-making. We first consider the possibility that LPs use interim rather than end-of-fund performance information in their selection decisions. In Table A.4, we report regressions that use interim IRRs for all funds, or alternatively, only for young GPs.²¹ We also use the IRR of the last fund to measure GP performance and report these results in the last column of Table A.4. In each of these cases, the coefficients on FirstTime and Young remain high relative to that on PastPerfHigh.

Next, we consider the possibility that missing performance (captured in the coefficient on PastPerfMiss) is non-random and alters inferences. For example, performance may be missing for funds with systematically poor or superior performance. We conduct two types of tests to assess this. First, we estimate a Heckman selection model with three variables in the first-stage. The first variable is calendar-year because missingness decreases over time due to improved database coverage. The second is an indicator variable (FOIAbility) which is equal to one for GPs who are subject to the Freedom of Information Act (FOIA). We consider a GP to be subject to FOIA reporting if it has received an allocation from a public pension system or a public university endowment eight years prior to the year of missingness.²² FOIAbility is a valid instrument because it does not directly influence the second stage equation, after controlling for an LP's or its peers' prior investment in a GP. The third variable is an indicator variable equal to one if eight years have passed since the GP's first fund. The coefficients on calendar year and FOIAbility are -0.006 (t -statistic = -3.07) and -0.718 (t -statistic = -23.76) respectively, and the regression has a pseudo- R^2 of 0.27. We report the coefficients of the second-stage regression in column (1) of Table A.5, and find that the Inverse Mills Ratio (Lambda) is statistically insignificant, and the other coefficients remain similar: coefficients on FirstTime and Young are 1.00% and 0.70% while the coefficient on PastPerfHigh is 1.02%.

²¹ The regressions in Table 4 require that at least five years elapse from the vintage year because there are strategic reporting issues with interim performance that could lead to misleading tests (Barber and Yasuda, 2017; Chung et al., 2012). On the other hand, interim performance could be important (Harris et al., 2023) and it is reasonable to ask whether interim performance changes the selection probabilities of young GPs.

²² We consider fund returns reliable if at least six years have passed and at the time of mandate allocation all past performance variables are lagged by two years, which adds up to eight years.

Second, we counterfactually assign funds with missing performance into the lowest or highest performance quartile, and then re-estimate the baseline regression in Table 4. In results reported in columns (2) and (3) of Table A.5, the coefficient on FirstTime is similar to that on PastPerfHigh.

4.4.2. First-time GPs' prior experiences and personal relationships

To deepen our understanding of the role of experience and human capital in the selection process for first-time GPs, we follow two paths. First, we hand-collect information on the prior employment history of up to five founding partners of the GP from the following sources: GP websites, Pitchbook, Bloomberg, LinkedIn, Crunchbase, and Tracxn. If a partner was previously employed by a private markets firm, or a non-private-markets firm that engaged in private market like activity (for example, the corporate venture capital arm of a non-financial firm), or a financial/real-estate firm, we label the first-time GP as Veteran GP. If not, or if we cannot find any employment history, we label the first-time GP as a Rookie GP.²³ We do not mean to imply that rookie GPs have no relevant experience but that they have no direct experience in private markets. Among the 2835 founding partners for whom we are able to gather data, 1149 have prior PE experience; 1005 have backgrounds in finance or real estate; 255 are entrepreneurs; 76 have legal background; 239 have experience in technology or biotechnology; and 111 have worked in healthcare. These categories align logically with firm specialization. For example, founding partners in real estate funds often have real estate backgrounds, while entrepreneurs are frequently involved in venture capital firms.

We re-estimate our baseline specification (column (3) from Table 4) after separating the FirstTime variable into its Rookie and Veteran components. Column (1) of Table 5 displays the regression results. The coefficients on the Rookie and Veteran variables are similar in magnitude, 0.94% and 1.01% respectively, and are comparable to that on PastPerfHigh (0.94%). We also investigate whether *where* the prior experience was acquired matters. We classify veteran GPs into two subcategories: if one of the founding partners acquired their experience through a private equity firm, we label them as VeteranPE, and if one of the founding partners acquired their experience from private equity firms in the largest size quartile we label them as VeteranLargePE. Our expectation is that the potential network and spillover effects from such GPs could also be larger, and LPs wishing to exploit such effects would be more likely to select them. Columns (2) and (3) indicate that VeteranPE has no additional effect relative to Veteran, while VeteranLargePE shows a small positive impact (0.11%, t -statistic = 1.91).

Second, we explore the role of personal relationships between LPs and first-time GPs. Given the regulatory framework around the solicitation of unregistered offerings, personal networks can play a potentially important role in fund raising (Howell et al., 2025). Although we have modeled the relaxation of constraints in the construction of the opportunity set in Section 4.2, the role of personal relationships in selection is also of independent interest. We create an indicator variable (PersRel) that is equal to one if a personal relationship is detected and zero otherwise. We introduce PersRel in column (4) of Table 5 along with its interaction with FirstTime. The coefficient on PersRel is 0.54% (t -statistic = 5.70), indicating that personal relationships are indeed positively related to selection for all GPs. The effect is even stronger for first-time GPs, as reflected in the interaction effect of 0.42% (t -statistic = 2.42).

²³ Some examples are helpful to understand the classification. For example, .406 Ventures was founded by Liam Donohue in 2005. Prior to that, Mr. Donohue was a principal at Foster Management, and then co-founded Arcadia Partners, both venture capital firms. Similarly, ACRE, a real-estate private equity firm has three founding partners, one of whom (Les Menkes) came with 24 years of experience with Morgan Stanley in the private wealth business. Both managers are labeled as Veteran GPs.

Table 5
Selection regressions with GP experience and personal connections.

	(1)	(2)	(3)	(4)
FirstTime	–	–	–	1.07 (10.36)
Rookie	0.94 (9.40)	0.95 (9.04)	0.95 (9.04)	–
Veteran	1.01 (9.82)	1.00 (9.81)	1.00 (9.80)	–
VeteranPE	–	–0.03 (–0.52)	–0.06 (–1.18)	–
VeteranPELarge	–	–	0.11 (1.91)	–
Young	0.67 (6.96)	0.67 (6.95)	0.67 (6.95)	0.78 (7.76)
PastPerfMiss	0.62 (5.44)	0.62 (5.42)	0.62 (5.43)	0.72 (6.31)
PastPerfQ2	0.22 (1.75)	0.21 (1.74)	0.21 (1.74)	0.30 (2.35)
PastPerfQ3	0.40 (3.39)	0.40 (3.38)	0.40 (3.39)	0.49 (4.05)
PastPerfHigh	0.94 (6.52)	0.93 (6.52)	0.94 (6.53)	1.02 (6.84)
PriorInv	9.71 (24.21)	9.71 (24.21)	9.71 (24.21)	9.69 (24.21)
PeerInv	0.57 (15.78)	0.57 (15.78)	0.57 (15.81)	0.57 (15.49)
Local	0.98 (17.09)	0.98 (17.09)	0.98 (17.09)	0.97 (17.03)
LnGPSize	0.50 (12.58)	0.50 (12.58)	0.50 (12.58)	0.49 (12.56)
PersRel	–	–	–	0.54 (5.70)
FirstTime × PersRel	–	–	–	0.42 (2.42)

This table presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in specification (3) of Table 4 with additional variables. Rookie is equal to one if this is the GP's first fund and no founding partner has prior private equity experience. Veteran is equal to one if this is the GP's first fund and at least one of the founding partners has prior experience in (a) a private markets firm, or (b) a non-private markets firm that engages in private market like activity (e.g., corporate venture capital), or (c) a financial and/or real estate firm. VeteranPE is equal to one if this is the GP's first fund and at least one of the founding partners had experience in a private equity firm. VeteranLargePE is equal to one if this is the GP's first fund and at least one of the founding partners had experience in a large private equity firm (where large is defined as a firm that ranks in the highest quartile of all private equity firms in terms of current assets under management in the year of fund raising). PersRel is an indicator variable equal to one if there is a personal relationship between the GPs and LPs, and zero otherwise. Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

4.5. Alternative opportunity sets and sampling variation

Selection probabilities are predicated on the curated exclusion restrictions used to generate the opportunity set. Some readers will undoubtedly disagree with our choices. For example, a reader who believes that GPs are willing to accept capital from any LP would naturally object to the access restrictions. Or, a reader who believes that GPs are willing to accept capital commitments of any size because they can scale funds appropriately would object to commitment size restrictions. In this section, we investigate alternative constructions of the opportunity set. In executing these alternative empirical designs, we re-estimate column (3) in Table 4 and report the results in Table 6. We also examine sampling variation, both in the set of GP-funds available to LPs, and in one case, in the set of LPs that we study. In

each of these constructions, the unconditional probability of selection varies, which means that all coefficients, including those on FirstTime and PastPerfHigh vary. We report the unconditional probability of selection in the last row for completeness, but as before, we focus on the coefficient on FirstTime relative to that on PastPerfHigh.

Removing access restrictions: In line with our earlier discussion of the possibility that private markets are a competitive business with an unrestricted supply of GPs and no access constraints, we remove all access restrictions described in 1(a) and 1(b). In column (1), the coefficient on FirstTime is 0.81%, substantially higher than on PastPerfHigh (0.46%).

Eliteness and rationing: A key criterion that we used to define elite GPs and therefore rationing was GP size. We consider two alternatives. The first considers the practitioner argument that only the top venture capital GPs engage in rationing, and other GPs are asset gatherers willing to take any LP's capital. In column (2), we apply the rationing restrictions only to VC funds. The coefficient on FirstTime is 0.86%, substantially higher than the coefficient on PastPerfHigh (0.53%). The second considers the case where elite GPs as those for whom at least 80% of the current fund's LPs were also investors in a prior fund (while requiring that the prior fund has at least 4 LPs in our database). The intuition is that elite GPs are those that largely only accept repeat investors. Column (3) shows that the coefficient on FirstTime (1.07%) is comparable to that on PastPerfHigh (1.05%).

Removing commitment size restrictions: In column (4), we remove the commitment size restrictions (2(a) and 2(b)) which effectively relaxes capacity constraints from both LP and GP perspectives. Here, too, the coefficient on FirstTime (1.06%) is comparable to PastPerfHigh (0.96%).

Personal relationships: In column (5), we relax all access restrictions (and the sole remaining commitment size constraint) when there is a personal relationship between LPs and GPs. In this specification, the coefficient on FirstTime is 1.04%, compared to 0.99% for PastPerfHigh.

Fund type restrictions: The empirical design described in Section 4.2 required that for small LPs, the opportunity set only includes fund types that the LP had invested in the past or current year. In column (6) we relax that restriction, allowing every LP, regardless of size, to choose from all GPs raising capital regardless of fund type. The coefficients on FirstTime is 0.81%, statistically indistinguishable from the coefficient on PastPerfHigh (0.86%). We also impose more stringent fund type restrictions by requiring that even large LPs only include those funds types in the opportunity set that they have historically or currently invested in. In column (7), this increases the coefficients on equivalent terms; the coefficient on FirstTime rises to 1.18%, again indistinguishable from the coefficient on PastPerfHigh (1.22%).

Definitions of large LPs: The separation of LPs into large and small subgroups plays a role in the construction of the opportunity set, and hence the selection probabilities. In the original design, we used the 90th percentile in the distribution of assets under management to identify these subgroups. In columns (8) and (9) we use the 70th and 80th percentile as breakpoints. In both of these specifications, the selection probabilities for first-time GPs are marginally higher than GPs in the highest past performance quartile. For example, in column (8), coefficient on FirstTime is 0.92%, compared to 0.83% on PastPerfHigh.

Private equity subsample: We estimate regressions in which we restrict the sample of both selected and non-selected funds to private equity

Table 6

GP selection regressions with varying opportunity sets and different samples.

	Alternative opportunity sets									Different samples		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
FirstTime	0.81 (13.59)	0.86 (12.71)	1.07 (10.28)	1.06 (10.59)	1.04 (10.31)	0.81 (9.42)	1.18 (10.19)	0.92 (10.40)	0.98 (10.47)	0.82 (6.97)	1.00 (10.96)	0.34 (2.30)
Young	0.53 (10.03)	0.58 (9.24)	0.77 (7.62)	0.76 (7.90)	0.73 (7.51)	0.58 (6.85)	0.85 (7.58)	0.64 (7.51)	0.69 (7.66)	0.52 (4.93)	0.75 (8.38)	0.19 (1.30)
PastPerfMiss	0.46 (7.60)	0.51 (7.16)	0.72 (6.17)	0.71 (6.40)	0.68 (6.09)	0.52 (5.47)	0.79 (6.17)	0.57 (5.98)	0.62 (6.11)	0.53 (4.25)	0.74 (7.11)	0.71 (3.87)
PastPerfQ2	0.18 (2.81)	0.21 (2.73)	0.31 (2.37)	0.30 (2.40)	0.29 (2.32)	0.23 (2.12)	0.34 (2.32)	0.28 (2.55)	0.29 (2.53)	0.20 (1.43)	0.39 (3.32)	0.06 (0.34)
PastPerfQ3	0.25 (4.28)	0.28 (4.05)	0.50 (4.17)	0.46 (4.01)	0.48 (4.05)	0.38 (3.59)	0.59 (4.31)	0.41 (4.08)	0.44 (4.14)	0.46 (3.08)	0.36 (3.20)	0.12 (0.55)
PastPerfHigh	0.46 (6.38)	0.53 (6.36)	1.05 (6.95)	0.96 (6.78)	0.99 (6.87)	0.86 (6.54)	1.22 (7.22)	0.83 (6.54)	0.89 (6.69)	0.94 (6.04)	1.18 (8.33)	0.47 (1.90)
PriorInv	10.76 (24.63)	10.65 (24.93)	9.69 (24.22)	9.85 (24.32)	9.78 (24.29)	9.60 (23.12)	9.67 (24.57)	9.81 (23.88)	9.79 (24.07)	11.62 (20.40)	9.02 (25.44)	15.70 (22.32)
PeerInv	0.25 (12.28)	0.28 (12.67)	0.59 (16.01)	0.54 (15.48)	0.56 (16.23)	0.47 (16.80)	0.65 (15.51)	0.43 (14.60)	0.47 (14.99)	0.42 (9.58)	0.54 (17.13)	1.61 (2.82)
Local	0.68 (17.31)	0.76 (15.70)	0.98 (17.08)	0.95 (17.12)	0.94 (16.91)	0.76 (17.89)	1.06 (17.03)	0.83 (16.88)	0.89 (16.90)	1.07 (14.75)	0.92 (17.99)	5.17 (12.96)
LnGPSize	0.21 (11.71)	0.25 (12.10)	0.50 (12.61)	0.47 (12.69)	0.49 (12.57)	0.41 (13.19)	0.56 (12.30)	0.44 (12.91)	0.46 (12.85)	0.47 (9.10)	0.46 (13.68)	0.72 (11.82)
# obs '000	8775	8078	6157	6282	6428	8337	5578	7405	6866	3542	7379	3479
Prob	0.70	0.76	1.00	0.98	0.96	0.74	1.10	0.83	0.90	0.97	0.95	0.77

This table presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in model (3) of Table 4, except that we use different opportunity sets in columns (1) to (9) and different samples in columns (10) to (12) as follows. Specification (1) removes all access restrictions, defined as 1(a) and 1(b). Specification (2) only imposes access restrictions for VC funds. Specification (3) adds an additional access restriction, where only large LPs are allowed to invest in elite GPs, who are defined as those whose prior fund had at least 80% repeated investments by LPs and had at least 4 LPs. Specification (4) removes restrictions associated with commitment sizes and the ratio of commitment sizes to target fund sizes (restrictions 2(a) and 2(b)). In specification (5) all access restrictions (and remaining commitment size constraints) are relaxed if there is a personal relationship. Specification (6) removes all fund type restrictions on the opportunity set (3(b)). Specification (7) requires that large LPs are also subject to fund type restrictions (3(a) is replaced with 3(b)). Specification (8) changes the definition of large LP to be those above the 70th percentile. Specification (9) changes the definition of large LP to be those above the 80th percentile. Specification (10) restricts the sample to commitments to venture capital, buyout and growth funds. Specification (11) includes fund-of-funds in both selected funds and the opportunity set. Specification (12) uses a sample of non-North-American LPs investing globally (includes international funds in both selected funds and the opportunity set). Fixed effects are based on LP $\times$ Year and FundRegion $\times$ FundType $\times$ VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The number of observations (in '000) and the unconditional probability of selection are reported in the last two rows. The sample in columns (1) to (11) consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

(buyout, venture, and growth). This provides points of comparison with much of the existing literature and also ensures that the selection probabilities for first-time and young GPs are not driven by non-equity funds. Column (10) reports that the coefficient on FirstTime remains quite high (0.82%) relative to that on PastPerfHigh (0.94%).

Fund of funds sample: Our primary tests exclude funds-of-funds because they operate as both LPs (investing in downstream funds) and GPs (receiving capital from LPs), and because our interest is in direct investment. Column (11) reports regressions in which funds-of-funds are included both as selected and non-selected funds. The coefficients on FirstTime and PastPerfHigh are 1.00% and 1.18% and statistically not different from each other, indicating that the inclusion of fund-of-funds does not change inferences.

International sample: Our tests focus on the North American market. However, we also estimate our baseline regression for a global sample of 26,765 allocations made by 6068 LPs from over 100 countries (excluding the US and Canada). The largest numbers of commitments come from the UK (4656), China (3768), and Japan (1970). We report the results from this sample in column (12) of Table 6. The coefficient on FirstTime is 0.34%, which is comparable to 0.47% for PastPerfHigh; the difference between the two coefficients is not statistically significant (*p*-value = 0.51). This exercise is informative not only as a robustness test, but also as evidence that our main conclusions from the North American sample likely extend to a global context.

4.6. Variation across investors

Our tests pool the selection decisions of all LPs. LP $\times$ Year fixed effects absorb variation across LPs but also obscure the role of economically interesting LP characteristics. We study several of these characteristics in Table 7 by re-estimating specification (3) in Table 4 for subsamples of LPs.

We hand-collect data on the presence of a formal investment committee (IC) for the LPs in our sample. The hand-collected data are derived from LP websites and public sources such as Institutional Investor. The first two columns of Table 7 examine the subsamples of LPs with and without ICs.²⁴ If there is no IC information available, we record the LP as not having an IC, but this reduces the differences in coefficients across the two regressions. In both regressions, the probability of selecting first-time and young GPs remains similar to that of GPs in the highest past performance quartile, implying that our results are not driven by the lack of an IC.

In the second set of columns, we estimate separate regressions for large and small LPs. In large LPs, the coefficient on FirstTime is almost twice that of the coefficient on PastPerfHigh (1.05% versus 0.55%). To the extent that LP size is related to scale economies in due diligence

²⁴ Andonov et al. (2018) and Binfarè et al. (2023) study aspects of the composition of ICs as they relate to performance for subsets of LPs, public pension funds and university endowments, respectively.

Table 7

Selection regressions for LPs with and without investment committees, LP size, LP experience, and LP type.

	By expertise (IC)		By size		By experience		By type			
	Yes	No	Large	Small	1	≥2	Corp. pensions	E & F	Publ. pensions	Misc.
FirstTime	1.26 (10.30)	0.77 (8.89)	1.05 (12.96)	1.08 (6.29)	1.10 (5.17)	1.06 (10.21)	0.89 (6.88)	0.83 (4.88)	1.62 (10.00)	0.70 (11.97)
Young	0.95 (8.06)	0.48 (5.70)	0.71 (10.64)	0.62 (3.58)	0.26 (1.26)	0.78 (7.77)	0.65 (5.13)	0.45 (2.66)	1.25 (8.09)	0.39 (7.40)
PastPerfMiss	0.86 (6.52)	0.47 (4.68)	0.52 (7.32)	0.58 (2.92)	0.62 (2.37)	0.71 (6.20)	0.56 (4.19)	0.56 (2.74)	1.02 (6.26)	0.36 (5.06)
PastPerfQ2	0.35 (2.38)	0.23 (1.97)	0.22 (2.85)	0.32 (1.32)	0.62 (1.88)	0.29 (2.29)	0.31 (2.01)	0.06 (0.25)	0.56 (3.05)	0.14 (1.91)
PastPerfQ3	0.56 (3.92)	0.38 (3.71)	0.31 (3.75)	0.94 (4.05)	0.93 (2.83)	0.48 (3.94)	0.61 (3.93)	0.48 (1.96)	0.71 (3.84)	0.25 (3.50)
PastPerfHigh	1.26 (7.30)	0.65 (5.00)	0.55 (6.03)	2.27 (7.21)	0.78 (2.78)	1.03 (6.85)	0.98 (4.97)	1.87 (5.58)	1.38 (6.56)	0.34 (4.18)
PriorInv	9.43 (24.17)	10.28 (22.07)	9.07 (22.82)	9.17 (18.89)	–	9.75 (24.32)	8.86 (18.27)	10.82 (20.89)	9.16 (22.15)	9.12 (21.13)
PeerInv	0.63 (14.47)	0.52 (15.09)	0.17 (7.18)	1.19 (17.13)	1.04 (7.74)	0.56 (16.07)	0.40 (9.24)	1.21 (12.39)	0.63 (11.37)	0.26 (9.34)
Local	1.00 (14.05)	0.94 (16.95)	0.72 (11.91)	1.16 (17.25)	2.52 (10.11)	0.90 (16.37)	0.57 (8.91)	0.93 (12.86)	1.56 (11.82)	0.72 (13.25)
LnGPSize	0.54 (12.11)	0.44 (12.93)	0.27 (11.62)	0.86 (13.17)	0.68 (9.53)	0.49 (12.49)	0.57 (12.38)	0.78 (11.11)	0.55 (11.43)	0.27 (11.87)
# obs '000	3653	2526	2793	3386	271	5908	1215	1505	1687	1774
Prob	1.11	0.83	0.83	1.13	0.96	1.00	0.97	1.08	1.26	0.69

This table presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in specification (3) of Table 4. Regressions are estimated separately for LPs with and without investment committees (ICs). LPs are classified as Large and Small based on the 90th percentile of the distribution of assets under management. LP experience is measured by counting the prior number of investments in private markets (1 and ≥2) at the time of the capital commitment. Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The number of observations (in '000) and the unconditional probability of selection are reported in the last two rows. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

and screening ability, this implies a considerable preference for first-time GPs. That does not appear to be the case for small LPs where the coefficient on FirstTime is 1.08% compared to 2.27% on PastPerfHigh.

The third set of columns considers experiential learning; an LP with prior investment experience in private markets may be more comfortable with the efficacy of its due diligence process than an LP making its first foray into private markets. This experience may be related to selection probabilities, especially for first-time and young GPs. We estimate regressions for the first year in which an LP makes commitments to any private market fund, and as a comparison group, we also show regressions for all subsequent commitment-years. The probability of LPs choosing first-time GPs in their first year is 1.10%, substantially higher than the probability of selecting a GP in the highest past performance quartile (0.78%). The results indicate that LPs are more likely to invest in first-time GPs during their initial entry into private markets than for their subsequent commitments. Nevertheless, the tendency to select first-time GPs remains strong even as LPs gain experience.

The last four columns contain regressions for corporate pension plans, endowments & foundations, public pension systems, and the miscellaneous category. There is variation in the relative selection probabilities that is interesting and suggestive of the pressures faced by different types of investors. In public pension systems, for example, liabilities are a function of contractual obligations and plan design, so there is limited flexibility; for these systems, the coefficient on FirstTime is 1.62% compared to 1.38% for PastPerfHigh. On the other hand, for Endowments & Foundations (which are less restricted in their future obligations), the coefficient on FirstTime is 0.83% compared to 1.87% on PastPerfHigh.

In summary, the results are best characterized as saying that LPs are not a monolith. While there is variation in selection probabilities across different subsets, in general, the tendency to select first-time GPs remains relatively high.

5. Explanations for commitments to first-time GPs

In this section, we explore potential explanations for why LPs have such a large propensity to invest in first-time GPs relative to GPs in the highest past performance quartile. There is one explanation that we can rule out based on existing tests. Some public pension funds create and manage Emerging Manager Programs that specifically allocate capital to GPs without a track record.²⁵ The fact that we observe relatively high selection probabilities for first-time GPs in the presence of LP × Year fixed effects indicates that the phenomenon is more widespread and involves deeper economic reasons. We proceed to other explanations below.

5.1. The role of expected performance

It is possible that LPs select first-time and young GPs because they expect their future performance to be high. To understand the role of expectations, we estimate expected performance using the methodology of Barber et al. (2021). In each year *t*, we run the following regression to estimate a fund's expected performance.

$$\begin{aligned}
 Perf_{j,\tau} = & \alpha_{0,t} + \alpha_{1,t} FirstTime_{j,v} + \alpha_{2,t} Young_{j,v} + \alpha_{3,t} PastPerfMiss_{j,v} \\
 & + \alpha'_{4,t} [PastPerfQ2_{j,v}, PastPerfQ3_{j,v}, PastPerfHigh_{j,v}] \\
 & + \alpha'_{5,t} FundType_{j,v} + e_{j,\tau},
 \end{aligned} \quad (3)$$

where *v* is the vintage year and *τ* is the year in which we observe fund *j*'s performance. To ensure that fund performance is observable at time *t*, we impose similar conditions as before, namely $\tau > (v +$

²⁵ See, for example, <https://www.calpers.ca.gov/page/investments/emerging-transition-manager-programs/emerging-manager-program> and <https://www.osc.state.ny.us/common-retirement-fund/emerging-manager>.

Table 8
Excess performance and selection.

	Panel A: Expected excess performance		Panel B: Realized excess performance		Panel C: Selection regression
	N	Mean	N	Mean	
FirstTime	2023	−1.65 (−3.23)	806	−1.08 (−1.91)	0.44 (9.85)
Young	2360	−1.99 (−10.05)	1257	−0.94 (−2.43)	0.24 (4.98)
PastPerfMiss	783	0.06 (0.12)	315	1.26 (1.57)	–
PastPerfLow	188	−3.74 (−4.05)	73	−3.19 (−2.34)	–
PastPerfQ2	396	5.01 (7.29)	173	−0.04 (−0.04)	−0.02 (−0.39)
PastPerfQ3	730	3.94 (9.18)	304	3.48 (1.46)	0.15 (1.82)
PastPerfHigh	861	4.45 (9.80)	318	2.63 (3.00)	0.27 (2.70)
PriorInv					8.93 (23.76)
PeerInv					0.52 (14.75)
Local					0.84 (16.64)
Ln GPSize					0.46 (11.85)

We calculate expected performance using Eq. (3). Using the fund-level expected performance, we compute excess average performance by subtracting the average performance of funds in the same type, region, and vintage year from a fund's expected performance. Panel A reports these averages. Panel B reports realized excess performance calculated as the difference between a fund's end-of-sample IRR and the average IRR of funds of the same type, geographic focus, and vintage year. In this panel, we require that more than five years elapse after the vintage year of the fund to be included in the sample. *t*-statistics for both panels are based on standard errors clustered at FundRegion × FundType × VintageYear appear in parentheses. Funds are placed into performance categories as defined in Table 2. Panel C presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in specification (3) of Table 4, except past return quartiles are replaced with expected return quartiles. Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The number of observations is 5,305,511 corresponding to an unconditional probability of selection of 0.94%. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

5) and $\tau \leq (t - 2)$. These rolling regressions are estimated with an expanding window, maximizing the use of past information but with no look-ahead bias. The fitted values represent expected performance. We then compute a fund's excess expected performance by subtracting the average expected performance of funds in the same type, region, and vintage year from the fund's expected performance.

Panel A of Table 8 reports the average excess expected performance for first-time GPs, young GPs, and GPs in each of the past performance quartiles. The average excess expected IRR of first-time GPs is −1.65% (*t*-statistic = −3.23). And the average expected excess IRR of young GPs is also negative (−1.99%). In contrast, the expected IRR of PastPerfHigh GPs is 4.45%. We also validate expected performance with fund-level realized performance. To do so, we calculate excess realized performance as the difference between a fund's end-of-sample IRR and the average IRR of funds of the same type, geographic focus, and vintage year. Panel B reports average excess realized performance of the funds from GPs in each of the past performance quartiles, as well as for the funds from first-time and young GPs. The fund-level excess realized IRR of first-time and young GPs is −1.08% and −0.94%, comparable to the regression based estimates.

The excess expected and realized performances are inconsistent with the notion that LPs select first-time and young GPs because of high expected future performance. To formally test whether expected returns

explains our findings, we re-estimate our baseline selection regression from Table 4 after including expected IRR in the regression. In Panel C, the coefficients on FirstTime and Young are 0.44% and 0.24%, respectively, vis-à-vis the coefficient on PastPerfHigh of 0.27%.

The distribution of expected and realized returns also permits us to speak to the role of fees as well as lottery preferences. Since both of these returns are measured net of fees, lower net returns for first-time GPs rule out the possibility that LPs prefer first-time GPs simply because they may charge lower fees. With regard to the argument the high coefficients on FirstTime and Young reflect a lottery preference, the skewness in realized IRRs for funds from first-time and young GPs is 4.8 and 1.8 respectively; funds from GPs in the third and highest performance quartiles have skewness of 9.9 and 7.7, suggesting that lottery preferences are an unlikely explanation.

5.2. Access to subsequent star funds

Practitioners sometimes argue that allocations to first-time GPs are a way for LPs to establish a foothold with GPs who subsequently become GPs in the highest quartile of performance and ration access.

Table 9
Commitments to first-time GPs and their subsequent funds.

Fund sequence #	# of commitments			Average realized subsequent performance
	Total	To PastPerfHigh GPs	With realized subsequent perf	
1	8479	–	4825	–0.60 (–0.75)
2	2604	20	1598	–0.30 (–0.32)
3	1315	116	802	–0.41 (–0.39)
4	734	136	355	1.51 (1.41)
5	381	101	167	3.67 (1.67)
6	271	74	89	0.47 (0.31)
7	117	52	50	–2.60 (–0.60)

The table reports the number of commitments and the average performance of LP investments in various vintages of funds raised by first-time GPs (the unit of observation is an LP investment in a fund). The first row reports the performance of investments in the first fund raised by a first-time GP. For the LPs who invested in the first-time fund of a GP, each subsequent row reports the average performance of investments in subsequent funds of the same GP. Performance is based on end-of-life excess IRR. We compute excess IRR by subtracting the average IRR of funds in the same type, region, and vintage year from a fund's IRR. We require that more than five years elapse after the vintage year of the fund to be included in the sample. *t*-statistics based on standard errors clustered at FundRegion $\times$ FundType $\times$ VintageYear appear in parentheses.

To assess the merit of this argument, we examine the sample of LPs who invest in a first-time GP, and follow these LPs' investments in subsequent funds of the same GP (i.e., reinvestment, conditional on investment in the first fund of the GP). Table 9 shows the number of re-investments in funds, up to the 7th fund. There are 8479 capital commitments to first-time GPs and 2604 commitments to the second fund of these GPs by the same LPs; extending the timeline, these LPs make 117 capital commitments to the 7th fund of these GPs. There is obvious survivorship bias built into the fund sequence in that ex post, only successful first-time GPs have the longevity to launch seven funds. There are 52 commitments to the 7th fund of surviving GPs who (ex post) are in the highest past performance quartile. These statistics suggest that relative to the large number of initial commitments in first-time GPs (8479), the probability of having an allocation to a fund that subsequently comes from a GP in the highest past performance quartile is low.

We also study the performance of commitments to funds in the sequence. The penultimate column of Table 9 shows the number of commitments with available data on realized performance. The last column shows the average excess IRR of such investments. Regardless of the fund sequence number, average reinvestment performance is indistinguishable from zero. Thus, there is little evidence to support the belief that investing in first-time GPs allows LPs, on average, to significantly benefit from access to future funds of these GPs.

5.3. Co-investment programs

It is possible that first-time GPs offer more co-investment opportunities to LPs in order to persuade them to become investors. Because coinvestment reduces total fees, this can be a viable channel (Braun et al., 2020; Fang et al., 2015). To investigate this, we acquire additional data from Preqin on which GPs allow coinvestment. Coverage is only available for about 15% of all funds, but in this limited sample, we observe very little difference in the percentage of funds that permit coinvestment across GPs in the past performance quartiles, first-time GPs, or young GPs. In fact, the fraction of first-time GPs offering coinvestment opportunities is 2% lower than that of GPs in the highest past performance quartile.

5.4. Demand and supply mechanisms

5.4.1. Motivating trends

A simple explanation for the propensity to invest in first-time GPs resides in the evolution of the demand and supply of private market funds. During our sample period, the portfolio weights of institutional investors in private markets increase steadily. Ivashina and Lerner (2018) report a large increase in allocations of global pension systems to private equity and real estate between 2008 and 2017, and argue that this increase reflects a willful shift to “perceived investment opportunities”. Preqin provides us with the total dollar investment in private markets for each LP-year. In Fig. 1, we report the sum of these allocations over time. We observe a steady increase in allocations, roughly doubling over the sample period. With such a large increase in allocations, market clearing requires either that the quantity supplied has to rise or that prices (fees) have to rise.

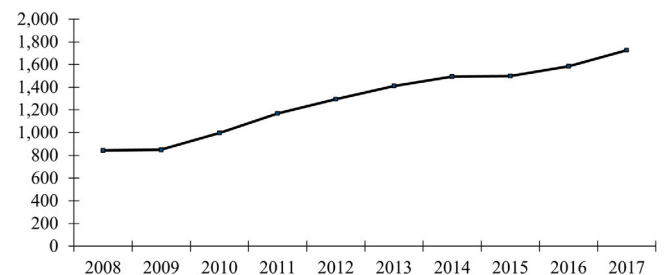


Fig. 1. PE allocations.

For each LP-year, Preqin reports the total dollar investment in private markets. We report the sum of these investments (in \$ billion) across all LPs over time.

A well-developed literature shows that high-performing incumbent GPs do not increase the fees or size of subsequent funds to fully clear the demand for their funds (Harris et al., 2023; Kaplan and Schoar, 2005). Hochberg et al. (2014) argue that incumbent LPs have bargaining power because they possess more knowledge about GP skill than outside investors, and use this power to limit fees. Marquez et al. (2015) argue that GPs may keep funds small to manipulate entrepreneur beliefs about their ability, thereby improving the flow of future investable firms (“deal flow”). There is also empirical evidence that organizational

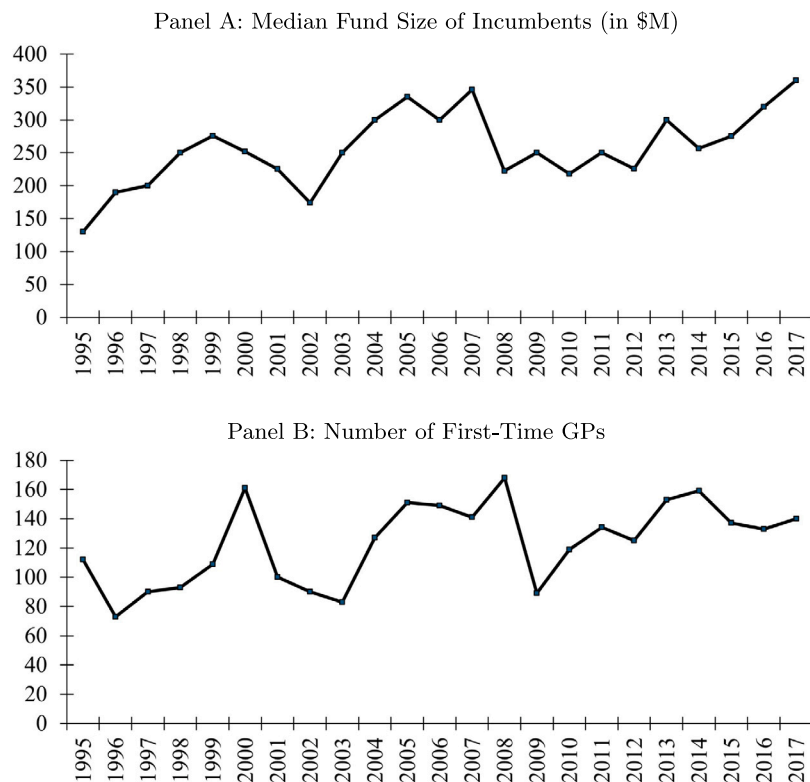


Fig. 2. Median fund size and the entry of new GPs.

Panel A shows the median fund size (in \$ million, rescaled by CPI in 2017 dollars) of incumbent funds in each vintage year. Panel B shows the number of new GPs (first-time GPs) in each vintage year.

diseconomies related to firm structure prevent the scalability of private equity funds (Lopez-de Silanes et al., 2015). These arguments imply that the increase in demand is likely satisfied by substitutes (alternative vehicles as described by Lerner et al., 2022), or by new entry.

To get a broad sense of these dynamics, we first tabulate the median fund size of incumbent GPs (that is, non-first-time GPs) and display the results in Panel A of Fig. 2. There are local peaks right before the dot-com and 2008 market breaks, but aside from that, there does not appear to be a significant increase in median fund size over the sample period. Panel B of Fig. 2 shows the number of new entrants. Again, there are local peaks in 2000 and 2007, but there is a secular increase in the number of first-time funds over time. The aggregate data, therefore, appear to be consistent with a simple story of demand met by the entry of first-time GPs.

5.4.2. The pressure to invest

The excess demand explanation implies that LPs facing high pressure to invest are more likely to invest in first-time GPs than LPs who do not face such pressure. Some of the results presented in Section 4.6 hint at this mechanism. For example, Table 7 shows that LPs with less discretionary liability structures (such as public and corporate pensions) are more likely to select first-time GPs; the coefficient on FirstTime for public and corporate pension systems are 1.62% and 0.89%, compared to 1.38% and 0.98% for PastPerfHigh respectively. In contrast, for endowments (which have more flexibility in spending and presumably face less pressure to invest) the coefficient on FirstTime is 0.83% compared to 1.87% for PastPerfHigh.

To measure this pressure to invest more directly, we combine two constructs. The first builds on the idea that LPs currently significantly underweight relative to their target allocations in a fund type face greater pressure to invest. The second employs the idea that LPs with higher growth in commitments in a fund type likely faced pressure

to invest. For the first construct, we obtain target weights from CEM Benchmarking but the data are limited to only 347 LPs in our sample. The second construct (dollar growth in capital commitments) has much broader coverage and is calculable for all LPs as long as an LP has capital commitments in consecutive years. Therefore, as a practical matter, combining the two constructs enhances the power of tests. We construct a measure of pressure-to-invest by defining an indicator variable (HighPres) equal to one for LPs in the highest dollar underweight quartile, or in the highest quartile of dollar growth in capital commitments.

Before using this variable in selection regressions, we first re-estimate the baseline regression from Table 4 to ensure that sampling variation does not alter prior results. Column (1) of Table 10 shows this baseline regression estimated for a sample that requires that HighPres be available. Consistent with previous regressions, the coefficient on FirstTime in this regression is 2.04%, compared to 2.15% on PastPerfHigh; the two coefficients, as in Table 4, are statistically indistinguishable from each other.

Column (2) of Table 10 contains coefficients from a regression that includes the HighPres variable and its interaction with FirstTime. The positive coefficient on HighPres indicates that LPs investing in multiple fund types are more likely to invest in the fund type where they face higher pressure. The coefficient on the interaction effect is 0.42% (t -statistic = 4.93), implying that LPs facing high pressure to invest are even more likely to invest in first-time GPs than LPs who face no such pressure to invest.²⁶

It is possible that LPs facing high pressure to invest have similar selection probabilities associated with other past performance categories.

²⁶ Since the regression contains LP-year fixed effects, the coefficient on HighPres is identified from LPs investing in multiple fund types but the coefficient on the interaction term is identified from all LPs.

Table 10
Selection regressions and the pressure to invest.

	(1)	(2)	(3)
FirstTime	2.04 (10.60)	1.91 (9.84)	1.69 (8.21)
Young	1.47 (7.66)	1.46 (7.65)	1.18 (5.85)
PastPerfMiss	1.39 (6.57)	1.39 (6.57)	1.18 (5.18)
PastPerfQ2	0.62 (2.51)	0.62 (2.53)	0.48 (1.90)
PastPerfQ3	1.18 (5.19)	1.18 (5.20)	1.02 (4.11)
PastPerfHigh	2.15 (7.54)	2.15 (7.54)	2.13 (6.74)
PriorInv	13.71 (28.17)	13.71 (28.12)	13.69 (28.07)
PeerInv	1.13 (16.12)	1.13 (16.09)	1.13 (16.10)
Local	1.77 (19.71)	1.77 (19.68)	1.77 (19.67)
LnGPSize	0.94 (13.47)	0.94 (13.50)	0.94 (13.54)
HighPres	–	0.94 (10.97)	0.42 (2.38)
FirstTime × HighPres	–	0.42 (4.93)	0.97 (5.54)
Young × HighPres	–	–	0.72 (4.46)
PastPerfMiss × HighPres	–	–	0.51 (2.91)
PastPerfQ2 × HighPres	–	–	0.32 (1.78)
PastPerfQ3 × HighPres	–	–	0.38 (1.59)
PastPerfHigh × HighPres	–	–	0.06 (0.23)

This table presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in specification (3) of Table 4, but including an additional variable. HighPres is an indicator variable equal to one if the LP faces pressure to invest in private markets (please see text for an exact definition). Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. Column (1) repeats the baseline specification (3) of Table 4 for the subsample for which we can construct the HighPres variable. All coefficients are multiplied by 100. The number of observations is 3,166,572 corresponding to unconditional probability of 1.84%. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

To investigate this, we interact HighPres with each of the performance categories, as well as FirstTime and Young. We report the results of such a regression in column (3) of Table 10. This richer specification reveals several interesting features of the selection mechanism.

First, the interaction effects fall (almost) monotonically across past performance groups. For example, the coefficient on the interaction of HighPres with FirstTime is 0.97%, that with Young is 0.72%, and falls to 0.06% for PastPerfHigh. Given the results in Table 8 (in which the expected and realized performance of FirstTime/Young GPs was lower than those of PastPerfHigh GPs), this pattern of selection probabilities suggests that LPs facing high pressure to invest navigate down the performance ladder. Interestingly, this monotonicity also helps rule out the possibility that HighPres captures a shift in preferences rather than true pressure to invest. For instance, LPs classified as HighPres might simply have experienced a shift in preferences toward a particular fund type after becoming overly optimistic about its expected returns (Begenau et al., 2024). If this preference-based explanation were true, such LPs

would invest more aggressively across all performance categories. The fact that the interaction effects decline monotonically suggests this is not the case.

Second, for LPs facing high pressure to invest, the implied selection probability for FirstTime is 2.66% (1.69 + 0.97) higher than the omitted group; the equivalent probability for PastPerfHigh is 2.19% (2.13 + 0.06). The difference between these two, 0.47% (*t*-statistic = 2.10) implies that LPs facing high pressure to invest are more likely to select first-time GPs than GPs in the highest past performance quartile.

Third, and perhaps even more importantly, for LPs under no pressure to invest, the incremental selection probability for first-time GPs is 1.69% (relative to the omitted group). This is substantially lower than the equivalent probability (2.13%) for PastPerfHigh, a difference of 0.44% (*t*-statistic = 1.97).

To summarize, LPs facing high pressure to invest show a strong tendency to invest first-time GPs relative to GPs in the highest past performance quartile. However, there is no such tendency for LPs that do not face high pressure to invest. This implies that the main mechanism driving the surprising tendency to invest in first-time GPs is the pressure to invest.

6. Conclusion

We study the selection decisions of 2692 North American LPs who make 61,602 capital commitments to private market funds between 1995 and 2019. Selection is of significance because of steadily increasing and large allocations by institutional investors (and even retail investors through the RIA and family office channels) into private markets. The asset class features long-term commitments of capital, high fees, considerable opacity, substantial due diligence and information costs, and the absence of passive investment options. All of these serve to highlight the importance of selection in private markets.

We use carefully curated opportunity sets to characterize the selection process and assess the economic importance of selection criterion. LPs chase performance, but perhaps most interestingly, also have large propensities to select first-time and young GPs. Indeed, the probability of selecting first-time GPs is as large as that of GPs in the highest past performance quartile. The most likely explanation for this is that excess demand from LPs is met by new GPs entering the industry. The results suggest that excess demand for exposure generates incentives for LPs to go down the quality ladder of fund supply. This raises the difficult question whether increasing allocations and the pressure to invest is beneficial for LPs from a portfolio perspective. Answering the question requires careful consideration of objective functions, expected liability structures, distributions frequencies, beliefs about asset class returns, variation in explicit costs (fees), and frictions associated with selection decisions. These are well beyond the focus of this paper but from our perspective, the frictional costs associated with selection are germane to these economically important decisions.

CRedit authorship contribution statement

Amit Goyal: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.
Sunil Wahal: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.
M. Deniz Yavuz: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors have nothing to disclose.

Appendix. Additional tables

Table A.1
Selection regressions with alternative fixed effects and clustering.

	(1)	(2)	(3)	(4)	(5)
FirstTime	–	1.46 (4.94)	1.22 (5.52)	1.02 (6.04)	1.06 (6.13)
Young	–	1.09 (4.12)	0.94 (4.41)	0.78 (4.86)	0.76 (4.80)
PastPerfMiss	–	0.97 (3.90)	0.87 (4.43)	0.73 (5.28)	0.71 (5.22)
PastPerfQ2	–	0.34 (1.77)	0.43 (2.57)	0.30 (1.95)	0.30 (2.03)
PastPerfQ3	–	0.45 (1.64)	0.54 (2.82)	0.50 (2.78)	0.49 (2.78)
PastPerfHigh	–	0.98 (2.67)	1.04 (3.33)	1.02 (4.05)	1.02 (4.06)
PriorInv	9.56 (25.46)	9.95 (8.16)	9.91 (8.10)	9.72 (8.03)	9.71 (8.05)
PeerInv	0.57 (16.16)	0.54 (6.81)	0.47 (6.83)	0.59 (6.62)	0.58 (6.56)
Local	0.99 (17.63)	0.98 (7.79)	0.97 (7.60)	0.98 (8.29)	0.98 (8.31)
LnGPSize	1.32 (10.87)	0.36 (3.74)	0.38 (3.89)	0.48 (4.10)	0.50 (4.17)
FE	LP × Year, FundRegion × FundType × VintageYear, GP × Year.	None	LP, FundType, VintageYear.	LP, FundRegion × FundType × VintageYear.	LP × Year, FundRegion × FundType × VintageYear.

This table presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in model (3) of Table 4, except that we use different fixed effects and clustering. Coefficients in column (5) of this table are the same as those in column (3) of Table 4. Fixed effects are denoted in the last row. We cluster standard errors in all columns at LP, FundType, and Year levels. *t*-statistics based on these standard errors are reported in parentheses. All coefficients are multiplied by 100. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

Table A.2
Selection regressions with TVPI.

	Clogit	OLS			
		(1)	(2)	(3)	Intermediary =
					Cons (4) PA (5)
FirstTime	1.38 (37.08)	1.02 (36.19)	0.95 (10.26)	0.95 (10.20)	0.99 (10.57)
Young	0.62 (18.69)	0.72 (26.81)	0.65 (7.12)	0.65 (7.12)	0.64 (7.02)
PastPerfMiss	0.55 (16.13)	0.69 (23.67)	0.60 (5.80)	0.61 (5.81)	0.60 (5.67)
PastPerfQ2	0.30 (8.56)	0.31 (9.09)	0.26 (2.16)	0.26 (2.17)	0.24 (2.07)
PastPerfQ3	0.23 (6.77)	0.44 (12.69)	0.43 (3.26)	0.43 (3.27)	0.37 (2.90)
PastPerfHigh	0.50 (14.92)	0.87 (22.59)	0.78 (5.53)	0.78 (5.54)	0.76 (5.41)
PriorInv	1.47 (60.87)	9.56 (89.80)	9.73 (24.30)	9.72 (24.27)	9.75 (24.35)
PeerInv	1.18 (66.76)	0.60 (47.51)	0.58 (16.10)	0.58 (16.03)	0.56 (15.65)
Local	0.81 (42.81)	0.99 (30.57)	0.98 (17.09)	0.97 (17.02)	0.97 (17.04)
LnGPSize	0.59 (91.31)	0.42 (69.71)	0.50 (12.64)	0.50 (12.64)	0.47 (12.16)
Intermediary	–	–	–	0.72 (12.19)	0.91 (12.25)

(continued on next page)

Table A.2 (continued).

	Clogit		OLS		
	(1)	(2)	(3)	Intermediary =	
				Cons (4)	PA (5)
FirstTime × Intermediary	–	–	–	0.16 (1.64)	–0.70 (–8.14)
FE	a	a	b	b	b

This table present results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in Table 4, except that we use TVPI (instead of IRR) to measure past performance. Fixed effects ‘a’ are based on LP × Year; ‘b’ are based on LP × Year and FundRegion × FundType × VintageYear; and ‘c’ are based on LP × Year, FundRegion × FundType × VintageYear, and GP × Year. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All OLS coefficients are multiplied by 100. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

Table A.3

Selection regressions with different failed funds.

	None (1)	Estimated & SPI (2)	SPI & Fund D (3)	Estimated & Fund D (4)
FirstTime	1.36 (9.30)	1.06 (10.33)	1.36 (9.69)	1.07 (9.96)
Young	0.95 (6.61)	0.76 (7.63)	0.96 (6.98)	0.77 (7.37)
PastPerfMiss	0.89 (5.32)	0.71 (6.16)	0.95 (6.07)	0.68 (5.65)
PastPerfQ2	0.45 (2.46)	0.30 (2.36)	0.44 (2.51)	0.31 (2.35)
PastPerfQ3	0.51 (3.05)	0.49 (4.07)	0.53 (3.33)	0.48 (3.90)
PastPerfHigh	1.07 (5.45)	1.02 (6.89)	1.10 (5.85)	1.01 (6.63)
PriorInv	10.79 (24.15)	9.71 (24.20)	10.76 (24.32)	9.75 (24.13)
PeerInv	0.96 (21.81)	0.58 (15.95)	0.94 (21.42)	0.58 (16.16)
Local	1.31 (19.20)	0.98 (17.08)	1.28 (18.77)	0.99 (17.32)
LnGPSize	0.61 (13.12)	0.50 (12.65)	0.58 (12.58)	0.51 (12.90)
# obs '000	4676	6175	4792	6068
Prob	1.32	1.00	1.29	1.01

This table present results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in model (3) of Table 4, except that we use different sets of failed funds (none; our estimation and from SPI data; SPI data and Fund D filings; and our estimation and from Fund D filings). Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The number of observations (in '000) and the unconditional probability of selection are reported in the last two rows. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

Table A.4

Selection regressions with interim performance.

	Interim		Last
	(1)	(2)	(3)
FirstTime	1.01 (9.35)	1.06 (10.39)	1.00 (7.57)
Young	0.70 (6.56)	0.75 (7.68)	0.68 (5.10)
Young × InterimPerfLow	–	–0.29 (–2.07)	–
Young × InterimPerfQ2	–	–0.37 (–2.09)	–
Young × InterimPerfQ3	–	0.10 (0.42)	–
Young × InterimPerfHigh	–	0.47 (2.20)	–
PastPerfMiss	0.54 (4.63)	0.71 (6.17)	0.59 (4.06)
PastPerfQ2	0.25 (2.00)	0.30 (2.36)	0.01 (0.06)
PastPerfQ3	0.50 (3.35)	0.49 (4.11)	0.42 (2.76)
PastPerfHigh	1.28 (8.42)	1.03 (6.90)	1.11 (6.24)
PriorInv	9.70 (24.38)	9.70 (24.22)	9.75 (24.50)
PeerInv	0.56 (15.68)	0.58 (16.16)	0.58 (15.76)
Local	1.00 (17.37)	0.98 (17.08)	1.00 (17.33)
LnGPSize	0.50 (12.77)	0.50 (12.67)	0.51 (12.88)

This table present results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in model (3) of Table 4. In column (1) we use interim performance for all GPs and place them into performance categories. In column (2) we use interim performance only for young GPs and their performance quartiles as additional variables. In column (3), we use the IRR of the last fund to measure GP performance. Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

Table A.5

Selection regressions with adjustments for missing past returns.

	Heckman	Robustness of PastPerfMiss	
	(1)	(2)	(3)
FirstTime	1.00 (8.87)	0.48 (8.35)	1.09 (10.42)
Young	0.70 (6.31)	0.18 (3.49)	0.78 (7.72)
PastPerfMiss	0.72 (6.19)	–	–
PastPerfQ2	0.30 (2.35)	–0.26 (–2.53)	0.30 (2.31)
PastPerfQ3	0.49 (4.07)	–0.08 (–0.74)	0.48 (3.92)
PastPerfHigh	1.02 (6.88)	0.45 (3.74)	0.85 (7.33)
PriorInv	9.71 (24.22)	9.70 (24.19)	9.74 (24.23)
PeerInv	0.58 (15.99)	0.56 (15.62)	0.59 (16.69)
Local	0.98 (17.07)	0.98 (17.05)	0.98 (17.05)
LnGPSize	0.50 (12.64)	0.50 (12.50)	0.50 (12.73)
Lambda	–0.18 (–1.26)	–	–

This table presents results of OLS regressions in which the dependent variable is equal to one if an LP commits capital to a fund of a GP, zero otherwise, similar to those in model (3) of Table 4. The Heckman model is described in the text. Column (1) reports the results from the second stage of the Heckman model. The two columns under the heading “Robustness of PastPerfMiss” explore the robustness to the variable PastPerfMiss. Column (2) assigns observations with PastPerfMiss to PastPerfLow. Column (3) assigns observations with PastPerfMiss to PastPerfHigh. Fixed effects are based on LP × Year and FundRegion × FundType × VintageYear. *t*-statistics are based on standard errors clustered using the same grouping as fixed effects. All coefficients are multiplied by 100. The sample consists of capital commitments by North American LPs between 1995 and 2019 to private market funds.

Data availability

Code and Pseudo Data (Reference data) (Mendeley Data)

References

- Andonov, A., 2024. Delegated investment management in alternative assets. *Rev. Corp. Finance Stud.* 13, 264–301.
- Andonov, A., Bonetti, M., Stefanescu, I., 2025. Choosing Pension Fund Investment Consultants. Technical Report, SSRN, <https://ssrn.com/abstract=4306217>.
- Andonov, A., Hochberg, Y.V., Rauh, J.D., 2018. Political representation and governance: Evidence from the investment decisions of public pension funds. *J. Finance* 73 (5), 2041–2086.
- Angrist, J.D., Pischke, J.-S., 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press, Princeton.
- Barber, B.M., Morse, A., Yasuda, A., 2021. Impact investing. *J. Financial Econ.* 139 (1), 162–185.
- Barber, B.M., Yasuda, A., 2017. Interim fund performance and fundraising in private equity. *J. Financial Econ.* 124 (1), 172–194.
- Barseghyan, L., Coughlin, M., Molinari, F., Teitelbaum, J.C., 2021. Heterogeneous choice sets and preferences. *Econometrica* 89 (5), 2015–2048.
- Begenau, J., Liang, P., Siriwardane, E., 2024. The Rise of Alternatives. Technical Report, SSRN, <https://ssrn.com/abstract=4940886>.
- Begenau, J., Robles-Garcia, C., Siriwardane, E., Wang, L., 2020. An Empirical Guide to Investor-Level Private Equity Data from Preqin. Technical Report, SSRN, <https://ssrn.com/abstract=3764895>.
- Begenau, J., Siriwardane, E., 2024. Fee variation in private equity. *J. Finance* 79 (2), 1199–1247.
- Binfare, M., Brown, G., Harris, R., Lundblad, C., 2023. How does human capital affect investing? Evidence from university endowments. *Rev. Finance* 27 (1), 143–188.
- Braun, R., Jenkinson, T., Schemmerl, C., 2020. Adverse selection and the performance of private equity Co-investments. *J. Financial Econ.* 136 (1), 44–62.
- Brown, G., Harris, R.S., Jenkinson, T., Kaplan, S.N., Robinson, D.T., 2015. What do Different Commercial Data Sets Tell Us About Private Equity Performance? Technical Report, SSRN, <https://ssrn.com/abstract=2701317>.
- Bursztyn, L., Ederer, F., Ferman, B., Yuchtman, N., 2014. Understanding mechanisms underlying peer effects: Evidence from a field experiment on financial decisions. *Econometrica* 82 (4), 1273–1301.
- Cain, M.D., McKeon, S.B., Solomon, S.D., 2020. Intermediation in private equity: The role of placement agents. *J. Financial Quant. Anal.* 55 (4), 1095–1116.
- Cassel, J., Lerner, J., Yimfor, E., 2022. Racial Diversity in Private Capital Fundraising. Technical Report, SSRN, <https://ssrn.com/abstract=4222385>.
- Cavagnaro, D.R., Sensoy, B.A., Wang, Y., Weisbach, M.S., 2019. Measuring institutional investors' skill at making private equity investments. *J. Finance* 74 (6), 3089–3134.
- Chade, H., Eeckhout, J., Smith, L., 2017. Sorting through search and matching models in economics. *J. Econ. Lit.* 55 (2), 493–544.
- Chiappori, P.-A., Salanié, B., 2016. The econometrics of matching models. *J. Econ. Lit.* 54 (3), 832–861.
- Chung, J.-W., Sensoy, B.A., Stern, L., Weisbach, M.S., 2012. Pay for performance from future fund flows: The case of private equity. *Rev. Financial Stud.* 25 (11), 3259–3304.
- Cochrane, J.H., 2005. The risk and return of venture capital. *J. Financial Econ.* 75 (1), 3–52.
- Crawford, G.S., Griffith, R., Iaria, A., 2021. A survey of preference estimation with unobserved choice set heterogeneity. *J. Econometrics* 222 (1), 4–43.
- Da Rin, M., Phalippou, L., 2017. The importance of size in private equity: Evidence from a survey of limited partners. *J. Financial Intermediation* 31, 64–76.
- Echenique, F., Oviedo, J., 2006. A theory of stability in many-to-many matching markets. *Theor. Econ.* 1, 233–273.
- Fang, L., Ivashina, V., Lerner, J., 2015. The disintermediation of financial markets: Direct investing in private equity. *J. Financial Econ.* 116 (1), 160–178.
- Fox, J.T., 2018. Estimating matching games with transfers. *Quant. Econ.* 9 (1), 1–38.
- Gaynor, M., Propper, C., Seiler, S., 2016. Free to choose? Reform, choice, and consideration sets in the english national health service. *Am. Econ. Rev.* 106 (11), 3521–3557.
- Goeree, M.S., 2008. Limited information and advertising in the U.S. personal computer industry. *Econometrica* 76 (5), 1017–1074.
- Goyal, A., Wahal, S., Yavuz, M.D., 2024. Choosing investment managers. *J. Financial Quant. Anal.* 59 (8), 3531–3563.
- Harris, R.S., Jenkinson, T., Kaplan, S.N., Stucke, R., 2023. Has persistence persisted in private equity? Evidence from buyout and venture capital funds. *J. Corp. Finance* 81, 102361.
- He, Y., Sinha, S., Sun, X., 2024. Identification and estimation in many-to-one two-sided matching without transfers. *Econometrica* 92 (3), 749–774.
- Hochberg, Y.V., Ljungqvist, A., Vissing-Jørgensen, A., 2014. Informational holdup and performance persistence in Venture capital. *Rev. Financial Stud.* 27 (1), 102–152.
- Hochberg, Y.V., Rauh, J.D., 2013. Local overweighting and underperformance: Evidence from limited partner private equity investments. *Rev. Financial Stud.* 26 (2), 403–451.
- Howell, S.T., Parker, D., Xu, T., 2025. Tyranny of the Personal Network: The Limits of Arm's Length Fundraising in Venture Capital. Technical Report, SSRN, <https://papers.ssrn.com/abstract=4990883>.
- Ivashina, V., Lerner, J., 2018. Looking for Alternatives: Pension Investments Around the World, 2008 to 2017. Technical Report, Harvard University.
- Kaplan, S.N., Schoar, A., 2005. Private equity performance: Returns, persistence, and capital flows. *J. Finance* 60 (4), 1791–1823.
- Katz, E., 2001. Bias in conditional and unconditional fixed effects logit estimation. *Political Anal.* 9 (4), 379–384.
- Lerner, J., Leamon, A., 2023. *Venture Capital, Private Equity, and the Financing of Entrepreneurship*, second ed. Wiley, Hoboken, NJ.
- Lerner, J., Mao, J., Schoar, A., Zhang, N.R., 2022. Investing outside the box: Evidence from alternative vehicles in private equity. *J. Financial Econ.* 143 (1), 359–380.
- Lerner, J., Schoar, A., 2004. The illiquidity puzzle: Theory and evidence from private equity. *J. Financial Econ.* 72 (1), 3–40.
- Lerner, J., Schoar, A., Wongsunwai, W., 2007. Smart institutions, foolish choices: The limited partner performance puzzle. *J. Finance* 62 (2), 731–764.
- Marquez, R., Nanda, V., Yavuz, M.D., 2015. Private equity fund returns and performance persistence. *Rev. Finance* 19 (5), 731–764.
- Martinez, J.V., Qian, Y., 2025. Selectivity, Favoritism, and Performance: The Role of Investment Consultants in Private Equity. Technical Report, SSRN, <https://ssrn.com/abstract=4322963>.
- McFadden, D., 1974. Conditional logit analysis of qualitative choice behavior. In: Zarembka, P. (Ed.), *Frontiers in Econometrics*. Academic Press, New York, pp. 105–142, Chapter 4.
- Phalippou, L., 2020. An inconvenient fact: Private equity returns and the billionaire factory. *J. Invest.* 30 (1), 11–38.
- Phalippou, L., Gottschalg, O., 2009. The performance of private equity funds. *Rev. Financial Stud.* 22 (4), 1747–1776.
- Preqin, 2016. Making the case for first-time funds. <https://docs.preqin.com/reports/Pr-eqin-Special-Report-Making-the-Case-for-First-Time-Funds-November-2016.pdf>.
- Robinson, D.T., Sensoy, B.A., 2013. Do private equity fund managers earn their fees? Compensation, ownership, and cash flow performance. *Rev. Financial Stud.* 26 (11), 2760–2797.

- Sensoy, B.A., Wang, Y., Weisbach, M.S., 2014. Limited partner performance and the maturing of the private equity industry. *J. Financial Econ.* 112 (3), 320–343.
- Lopez-de Silanes, F., Phalippou, L., Gottschalg, O., 2015. Giants at the gate: Investment returns and diseconomies of scale in private equity. *J. Financial Quant. Anal.* 50 (3), 377–411.
- Sørensen, M., 2007. How smart is smart money? A two-sided matching model of venture capital. *J. Finance* 62 (6), 2725–2762.
- Stammann, A., Heiss, F., McFadden, D., 2016. Estimating Fixed Effects Logit Models with Large Panel Data. *VfS Annual Conference 2016 (Augsburg): Demographic Change*, Verein für Socialpolitik/German Economic Association.
- Swensen, D.F., 2000. *Pioneering Portfolio Management: An Unconventional Approach to Institutional Investment*. Free Press, New York.